

ATIYAH CLASSES AND ELLIPTICITY OF OKA MANIFOLDS

YUTA KUSAKABE AND SHIN-ICHI MATSUMURA

ABSTRACT. We prove that every weakly pseudoconvex Oka manifold admitting a positive line bundle is elliptic in the sense of Gromov, thereby proving Gromov's ellipticity conjecture.

The proof establishes a cohomological construction of local sprays. Given a bundle morphism $\varphi: E \rightarrow T_X$, we introduce quadratic φ -vector fields on E whose flows yield local sprays with fibre derivative φ . Their existence is characterized by the vanishing of the symmetrized φ -Atiyah class of E . This gives a local dominating spray on every weakly pseudoconvex manifold admitting a positive line bundle, without any Oka assumption.

The positivity hypothesis is essential even for local existence: blow-ups at a single point of complex tori of algebraic dimension zero, and Kummer surfaces of algebraic dimension zero, admit no local dominating spray. The torus examples give compact Kähler Oka manifolds that are not elliptic and show that ellipticity is not preserved under blowing up, and that it is neither open nor closed in holomorphic families of compact Oka manifolds. Combined with the work of Xie and Zhao, the Kummer examples yield Oka K3 surfaces that are not elliptic.

Under the additional Oka assumption, we globalize the local dominating sprays constructed above. The proof combines Oka approximation with a gluing argument based on weighted L^2 estimates for the $\bar{\partial}$ -equation.

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1. INTRODUCTION

We investigate the relation between the Oka property and ellipticity through the existence and globalization of local dominating sprays. Our main result is the following.

Theorem 1.1 (Gromov’s conjecture for weakly pseudoconvex manifolds). *Every weakly pseudoconvex Oka manifold admitting a positive line bundle is elliptic in the sense of Gromov.*

A complex manifold X is called *Oka* if every continuous map from a Stein manifold to X satisfies the homotopy principle; that is, it can be deformed into a holomorphic map, with approximation and interpolation (see [8, Chap. 5] for the precise definition, and [7] and [8, Sect. 5.4] for equivalent formulations). A complex manifold X is called *elliptic* if a vector bundle $p: E \rightarrow X$ admits a dominating spray; that is, a holomorphic map $s: E \rightarrow X$ from the total space of E such that $s(0_x) = x$ and $(ds)_{0_x}(E_x) = T_{X,x}$ for every $x \in X$.

Gromov proved that every elliptic manifold is Oka [11] (see [8, Chap. 5] for a proof and [8, Sect. 6.4] for examples). He stated, without proof, that the converse holds for holomorphically convex manifolds admitting a positive line bundle [11, 3.2.A’]. We refer to this as *Gromov’s conjecture*. He also observed that the converse holds for Stein manifolds [11, 3.2.A]. The converse fails without additional assumptions: the first author found Oka manifolds that are not elliptic [14]. For instance, if $n \geq 2$, the complement $\mathbb{C}^n \setminus K$ of a compact polynomially convex set K is Oka [15, Theorem 1.6], whereas it is not elliptic if $n \geq 3$ and K has nonempty interior [2]. All these examples are noncompact. Recently, Forstnerič and Lárusson settled the projective case [10]. Theorem 1.1 unifies and extends these results, proving Gromov’s conjecture for weakly pseudoconvex manifolds admitting a positive line bundle. The class of weakly pseudoconvex manifolds includes Stein, projective, and holomorphically convex manifolds and arises naturally in several complex variables.

The positivity hypothesis cannot be omitted, even among compact Kähler Oka manifolds. Our second main result shows that this obstruction already occurs at the level of local dominating sprays.

Theorem 1.2 (Compact Kähler manifolds without local dominating sprays; see Examples 2.9 and 2.10).

- (1) *Let $\mathbb{T}$ be a complex torus of algebraic dimension 0. Then the blow-up $\pi: X := \text{Bl}_{t_0}(\mathbb{T}) \rightarrow \mathbb{T}$ at a point $t_0 \in \mathbb{T}$ admits no local dominating spray.*
- (2) *Every Kummer surface X of algebraic dimension 0 admits no local dominating spray.*

Here a local dominating spray is a dominating spray $s: U \rightarrow X$ defined only on an open neighbourhood U of the zero section.

The manifold X in the first assertion is Oka by [8, Corollary 6.4.13] (see also Buzzard’s theorem on tame sets [5]). Thus, it gives the first example of a compact Kähler Oka manifold that is not elliptic, answering a question of Forstnerič and Lárusson [10]. The same example shows that ellipticity need not be preserved under blowing up, even among compact Kähler manifolds. Moreover, varying the torus in holomorphic families shows that

ellipticity is neither an open nor a closed condition in holomorphic families of compact Oka manifolds (Remark 2.12); recall that the Oka property itself is not closed in families [9].

Combined with the work of Xie and Zhao [23], the second assertion yields Oka K3 surfaces that are not elliptic. It also gives another instance of the failure of openness: there is a holomorphic family of Kummer surfaces whose central fibre is the elliptic Fermat quartic, whereas all fibres outside a countable set of parameters are not elliptic.

We divide the construction of dominating sprays in the proof of Theorem 1.1 into two problems:

- (1) *Local existence.* When does a complex manifold X , without any Oka assumption, admit a vector bundle $E \rightarrow X$ carrying a local dominating spray?
- (2) *Globalization.* Given a local dominating spray on a weakly pseudoconvex Oka manifold, under what additional conditions can we construct a global dominating spray?

A local dominating spray $s: U \rightarrow X$ on a vector bundle $E \rightarrow X$ induces a surjective bundle morphism $E \rightarrow T_X$ through its fibre derivative. This leads to the following question:

- Given a surjective bundle morphism $\varphi: E \rightarrow T_X$, when can we integrate φ to a local spray on E whose fibre derivative is φ ?

We develop a cohomological approach to the local existence problem. For this purpose, we introduce *quadratic φ -vector fields* on the total space of E (see Definition 2.2 and Remark 2.3). Their flows integrate φ to a local spray, called an *exponential φ -spray* (see Proposition 2.4).

The existence of these vector fields is characterized by the following cohomological criterion using the Atiyah class [3].

Theorem 1.3 (Cohomological criterion for quadratic φ -vector fields, Theorem 2.6). *The following conditions are equivalent:*

- (1) *The bundle E admits a quadratic φ -vector field.*
- (2) *The symmetrized φ -Atiyah class $\alpha_\varphi(E)$ vanishes.*

Here $\alpha_\varphi(E)$ is defined in terms of the Atiyah class $a(E)$ of E (see Definition 2.5). The theorem identifies the obstruction to constructing quadratic φ -vector fields and gives a sufficient condition for integrating a prescribed bundle morphism into a local spray. When φ is surjective, the resulting spray is dominating. Applying this criterion, we construct a local dominating spray on every weakly pseudoconvex manifold admitting a positive line bundle (Theorem 4.1). This construction does not require the Oka property.

The Oka property enters in the globalization step, expressed by the following theorem.

Theorem 1.4. *Let X be a weakly pseudoconvex Oka manifold, and let $B \rightarrow X$ be a positive line bundle such that $K_X^{-1} \otimes B^{\otimes k}$ admits a smooth Hermitian metric with semipositive curvature for some integer $k \geq 0$. If $E = (B^\vee)^{\oplus N}$ admits, for some $N \geq 1$, a local dominating spray s , then E admits a global dominating spray that agrees with s to order 2 along the zero section. In particular, X is elliptic.*

Here two holomorphic maps *agree to order r* along a submanifold A if, in local coordinates on the target, all their partial derivatives of order $< r$ agree at the points of A . Theorem 1.4 follows from a jet extension theorem for maps from the total space of a negative vector bundle into an Oka manifold, with interpolation along the zero section (see Theorem 3.1).

In the projective case, Forstnerič and Lárusson used Stopar's Oka principle on 1-convex domains [20, Theorem 3.3] (cf. [18]). In our setting, the zero section may be noncompact, so E need not be 1-convex and these results do not apply in general. The domination off the zero section required in Theorem 3.1 is provided by Lemma 3.9, the analogue of [10, Lemma 2.2]. The proof of Theorem 3.1 follows the local-to-global scheme of the Oka principle for Stein sources [8, Chap. 5], with a gluing argument based on weighted L^2 estimates for the $\bar{\partial}$ -equation that preserves the prescribed jets along the zero section. Here we treat the jet extension theorem only in the form needed to globalize sprays. The method also extends to maps from general weakly pseudoconvex manifolds into Oka manifolds; we will address this elsewhere.

Organization. Section 2 treats local sprays: Section 2.1 establishes the cohomological criterion, and Section 2.2 proves the nonexistence results above. Section 3 is devoted to globalization: Section 3.1 proves a splitting lemma on the weakly pseudoconvex total space of a negative bundle, and Section 3.2 proves Theorem 3.1. Section 4 proves Theorem 4.1 and Theorems 1.1 and 1.4.

2. EXISTENCE OF LOCAL DOMINATING SPRAYS

2.1. Quadratic φ -vector fields and a cohomological integration criterion. In this subsection, for a given bundle morphism $\varphi: E \rightarrow T_X$, we introduce quadratic φ -vector fields and show, by considering their flows, that $\varphi: E \rightarrow T_X$ can be integrated to local sprays, called exponential φ -sprays (see Proposition 2.4). We also give a cohomological criterion for the existence of quadratic φ -vector fields (see Theorem 2.6).

Throughout this subsection, we work in the following setup.

Setup 2.1. Let X be a complex manifold of dimension n , let $p: E \rightarrow X$ be a vector bundle of rank r , and let $\varphi: E \rightarrow T_X$ be a bundle morphism, not necessarily surjective. We write 0_x for the zero vector in E_x and $0_X \subset E$ for the image of the zero section. For $t \in \mathbb{C}$, let $m_t: E \rightarrow E$ denote fibrewise multiplication by t .

We first define quadratic φ -vector fields.

Definition 2.2. A *quadratic φ -vector field* is a vector field Z on the total space of E satisfying the following conditions:

- $dp_v(Z(v)) = \varphi_x(v) \quad (x \in X, v \in E_x).$
- $Z(tv) = t(dm_t)_v Z(v) \quad (v \in E, t \in \mathbb{C}).$

Here $(dm_t)_v: T_v E \rightarrow T_{tv} E$ denotes the differential of m_t at $v \in E$.

Remark 2.3. The definition is modelled on sprays in differential geometry in the sense of Ambrose, Palais and Singer [1]. Indeed, for $E = T_X$ and $\varphi = \text{id}_{T_X}$, the first condition

says that Z is a second-order vector field, and the second one is the homogeneity condition in [1]; thus a quadratic id_{T_X} -vector field is a spray on X in their sense. The exponential φ -spray in Proposition 2.4 below corresponds to the exponential map of such a spray.

We show that a quadratic φ -vector field on E integrates $\varphi: E \rightarrow T_X$ to a local spray.

Proposition 2.4 (Integration of quadratic φ -vector fields). *Every quadratic φ -vector field Z on E is integrable; that is, there exist an open neighbourhood $U \subset E$ of 0_X and a holomorphic map $s: U \rightarrow X$ such that*

$$s(0_x) = x, \quad \left. \frac{d}{d\lambda} \right|_{\lambda=0} s(\lambda v) = \varphi_x(v) \quad (x \in X, v \in E_x).$$

More precisely, such a map $s: U \rightarrow X$ is given by $s = p \circ \text{Fl}_1^Z$, where Fl_t^Z denotes the flow of Z ; we call it the exponential φ -spray of Z . In particular, if φ is surjective, then s is a local dominating spray.

Proof. Step 1. Local expression of quadratic φ -vector fields. We first express the defining conditions of quadratic φ -vector fields in local coordinates. Take an open set $V \subset X$ and a frame $e = (e_1, \dots, e_r)$ of $E|_V$, regarded as a row vector. This frame defines the trivialization

$$V \times \mathbb{C}^r \longrightarrow E|_V, \quad (x, w) \longmapsto e(x)w$$

and the natural isomorphism

$$T_{e(x)w}E \cong T_x X \oplus \mathbb{C}^r.$$

With respect to this decomposition, the restriction of Z to $E|_V$ can be written as

$$Z(x, w) = (A(x, w), B(x, w)),$$

where $A(x, w) \in T_x X$ is its base component and $B(x, w) \in \mathbb{C}^r$ is its fibre component.

Since $p(x, w) = x$ in this trivialization, we have $dp_{(x, w)}(c, d) = c$. Thus, for $v = e(x)w$, the first condition $dp_v(Z(v)) = \varphi_x(v)$ is equivalent to

$$(1) \quad A(x, w) = \varphi_x(e(x)w).$$

In particular, the base component is uniquely determined by φ and is linear in w .

The fibrewise multiplication m_t is given by $m_t(x, w) = (x, tw)$, so its differential is $(dm_t)_{(x, w)}(c, d) = (c, td)$. Consequently, we obtain

$$t(dm_t)_{(x, w)}Z(x, w) = (tA(x, w), t^2B(x, w)).$$

Thus, the second condition $Z(tv) = t(dm_t)_v Z(v)$ is equivalent to

$$A(x, tw) = tA(x, w) \text{ and } B(x, tw) = t^2B(x, w).$$

The first equality already follows from the expression (1) for A . The second requires the fibre component to be homogeneous of degree two: one factor of t comes from the differential dm_t , and the other from the scalar multiplying this differential. This homogeneity is equivalent to the existence of a symmetric bilinear map $\gamma(x): \mathbb{C}^r \times \mathbb{C}^r \rightarrow \mathbb{C}^r$, depending holomorphically on x , such that

$$(2) \quad B(x, w) = -\gamma(x)(w, w).$$

The minus sign is a convention chosen for the subsequent calculations.

Step 2. Construction of the local spray by the flow. Consider the local flow Fl_t^Z of Z , characterized by

$$\frac{d}{dt} \text{Fl}_t^Z(v) = Z(\text{Fl}_t^Z(v)), \quad \text{Fl}_0^Z(v) = v.$$

The local existence of the flow and its holomorphic dependence on the initial point v follow from the theory of holomorphic ordinary differential equations.

We show that the flow is defined for $0 \leq t \leq 1$ on a sufficiently small neighbourhood of the zero section. Fix $x \in X$. There exist an open neighbourhood W of 0_x and $\varepsilon > 0$ such that the local flow is defined for initial points in W and $|t| < \varepsilon$. The local expressions (1) and (2) show that $Z(0_x) = 0$ for every $x \in X$. Hence the solution starting at 0_x is constant. Thus, for an integer $N \in \mathbb{Z}_+$ with $1/N < \varepsilon$, the holomorphic map $\text{Fl}_{1/N}^Z$ fixes 0_x . After shrinking the neighbourhood of 0_x , its N -fold composition is well defined. By uniqueness of solutions, this composition is the time-one map Fl_1^Z . Applying this construction at every point of the zero section and using uniqueness on overlaps gives a holomorphic map

$$\text{Fl}_1^Z: U \longrightarrow E$$

on an open neighbourhood U of 0_X . Since the neighbourhood is allowed to depend on x .

We show that the holomorphic map

$$s := p \circ \text{Fl}_1^Z: U \rightarrow X$$

is a local spray whose fibre derivative coincides with φ . Since the flow fixes the zero section, we have $s(0_x) = x$. It remains to compute its fibre derivative. Fix $x \in V$ and $v = e(x)w \in E_x$. For $\lambda \in \mathbb{C}$ sufficiently close to zero, the orbit starting at λv remains in this coordinate neighbourhood for $0 \leq t \leq 1$. We write

$$\text{Fl}_t^Z(\lambda v) = (z(t, \lambda), w(t, \lambda)).$$

By (1) and (2), the flow equations are

$$(3) \quad \begin{aligned} \frac{\partial z}{\partial t} &= \varphi(z(t, \lambda))w(t, \lambda), \\ \frac{\partial w}{\partial t} &= -\gamma(z(t, \lambda))(w(t, \lambda), w(t, \lambda)), \end{aligned}$$

with initial conditions

$$z(0, \lambda) = z(x), \quad w(0, \lambda) = \lambda w.$$

Here $\varphi(z)$ denotes the matrix of φ with respect to the frame e and the base coordinates z , and $\gamma(z)$ denotes the corresponding local coefficient expression of the quadratic term.

Differentiate the second equation in (3) with respect to λ and set $\lambda = 0$. The terms obtained by differentiating the coefficients of γ contain two factors $w(t, 0)$, while those obtained by differentiating either argument of γ contain one such factor. All these terms vanish, so we obtain

$$\frac{d}{dt} \left(\frac{\partial w(t, \lambda)}{\partial \lambda} \Big|_{\lambda=0} \right) = 0.$$

Using the initial condition, we obtain

$$\left. \frac{\partial w(t, \lambda)}{\partial \lambda} \right|_{\lambda=0} = w \quad (0 \leq t \leq 1).$$

Next, differentiate the first equation in (3) with respect to λ . The term obtained by differentiating the coefficients of φ is multiplied by $w(t, 0) = 0$ and therefore vanishes. The remaining term is $\varphi(z(x))w$, the coordinate expression of $\varphi_x(v)$. Thus, identifying tangent vectors with their coordinate expressions, we obtain

$$\frac{d}{dt} \left(\left. \frac{\partial z(t, \lambda)}{\partial \lambda} \right|_{\lambda=0} \right) = \varphi_x(v).$$

Since $z(0, \lambda)$ is independent of λ , integration gives

$$\left. \frac{\partial z(t, \lambda)}{\partial \lambda} \right|_{\lambda=0} = t \varphi_x(v).$$

Evaluating at $t = 1$ yields

$$\left. \frac{d}{d\lambda} \right|_{\lambda=0} s(\lambda v) = (dp)_{0_x}((d\mathrm{Fl}_1^Z)_{0_x}(0, v)) = (dp)_{0_x}(\varphi_x(v), v) = \varphi_x(v)$$

as required. $\square$

We now define the cohomology class appearing in the criterion.

Definition 2.5. The *symmetrized φ -Atiyah class* of E is defined by

$$\alpha_\varphi(E) := \mathrm{Sym}(\varphi^* a(E)) \in H^1(X, S^2 E^\vee \otimes E),$$

where $a(E) \in H^1(X, \Omega_X^1 \otimes \mathrm{End} E)$ is the Atiyah class of E and φ^* denotes the map on cohomology induced by $\varphi^*: \Omega_X^1 \rightarrow E^\vee$, so that

$$\varphi^* a(E) \in H^1(X, E^\vee \otimes \mathrm{End} E) \cong H^1(X, E^\vee \otimes E^\vee \otimes E).$$

The morphism Sym symmetrizes the two factors $E^\vee$: for an E -valued bilinear form B , we set

$$\mathrm{Sym} B(u, v) := \frac{1}{2}(B(u, v) + B(v, u)).$$

Theorem 2.6 (Cohomological criterion for quadratic φ -vector fields). *The following conditions are equivalent:*

- (1) *The bundle E admits a quadratic φ -vector field.*
- (2) *The symmetrized φ -Atiyah class $\alpha_\varphi(E)$ vanishes.*

Proof. Every local trivialization of E yields a local quadratic φ -vector field. We show that local quadratic φ -vector fields can be chosen so that they glue together precisely when a cocycle representing $\alpha_\varphi(E)$ is a coboundary.

Step 1. A cocycle representing $\alpha_\varphi(E)$. Let $\{V_i\}$ be a locally finite open cover of X by Stein open sets, and let $e_i := (e_{i,1}, \dots, e_{i,r})$ be a frame of $E|_{V_i}$. Let $\{g_{ij}\}$ denote the transition matrices of E , defined by $e_i = e_j g_{ij}$. The fibre coordinates $w_i: E|_{V_i} \rightarrow \mathbb{C}^r$ are the column vectors defined by

$$v = e_i(x) \cdot w_i(v) \quad (v \in E_x, x \in V_i).$$

Then

$$(4) \quad w_j = (g_{ij} \circ p)w_i \quad \text{on } E|_{V_i \cap V_j}.$$

Let ∇^i be the holomorphic connection on $E|_{V_i}$ that differentiates the coefficients of a section in the frame e_i :

$$\nabla^i(e_i u_i) = e_i du_i.$$

On $V_{ij} := V_i \cap V_j$, the Leibniz rule shows that $A_{ij} := \nabla^j - \nabla^i$ is $\mathcal{O}_X$ -linear, so $A_{ij} \in H^0(V_{ij}, \Omega_X^1 \otimes \text{End}(E))$. With our sign convention, the cocycle $\{A_{ij}\}$ represents the Atiyah class $a(E)$. For a section $u = e_i u_i = e_j u_j$, we have $u_j = g_{ij} u_i$. Consequently,

$$\nabla^j u = e_j d(g_{ij} u_i) = e_i (du_i + g_{ij}^{-1} dg_{ij} u_i).$$

Subtracting $\nabla^i u = e_i du_i$, we obtain

$$(5) \quad A_{ij} = g_{ij}^{-1} dg_{ij},$$

where we identify A_{ij} with its matrix of holomorphic 1-forms in the frame e_i .

Pulling back by φ and symmetrizing, we obtain the cocycle

$$C_{ij}(u, v) := \frac{1}{2} (A_{ij}(\varphi(u))v + A_{ij}(\varphi(v))u) \quad (u, v \in E_x, x \in V_{ij}).$$

By definition, the symmetrized φ -Atiyah class $\alpha_\varphi(E) \in H^1(X, S^2 E^\vee \otimes E)$ is represented by this cocycle. Here $A_{ij}(\varphi(u))$ denotes the endomorphism of E_x obtained by evaluating the 1-form factor of A_{ij} on $\varphi(u)$. In particular, $C_{ij}(v, v) = A_{ij}(\varphi(v))v$.

Step 2. A gluing condition for quadratic φ -vector fields. Given a section $\Gamma_i \in H^0(V_i, S^2 E^\vee \otimes E)$, we define a symmetric bilinear map $\gamma_i(x): \mathbb{C}^r \times \mathbb{C}^r \rightarrow \mathbb{C}^r$ by

$$\Gamma_{i,x}(e_i(x)a, e_i(x)b) = e_i(x)\gamma_i(x)(a, b).$$

Using γ_i , we then define a local quadratic φ -vector field Z_i on $E|_{V_i}$ by (1) and (2).

We show that the vector fields Z_i glue together to a global vector field on E if and only if the sections Γ_i satisfy

$$(6) \quad \Gamma_i - \Gamma_j = C_{ij}$$

on every overlap.

To verify this, we compare the actions of Z_i and Z_j on holomorphic functions. By (1), Z_i and Z_j agree on functions pulled back from X . It remains to compare their actions on the components of w_j . Fix $x \in V_i \cap V_j$ and $v \in E_x$. Applying Z_i to (4) and using the Leibniz rule, we obtain

$$(7) \quad \begin{aligned} Z_i(w_j)(v) &= Z_i(g_{ij} \circ p)(v)w_i(v) + g_{ij}(x)Z_i(w_i)(v) \\ &= (dg_{ij})_x(\varphi_x(v))w_i(v) - g_{ij}(x)\gamma_i(x)(w_i(v), w_i(v)) \end{aligned}$$

by (1) and (2). Here the vector field acts componentwise on the matrix $g_{ij} \circ p$ and on the coordinate maps w_i, w_j . On the other hand, (2) gives $Z_j(w_j)(v) = -\gamma_j(x)(w_j(v), w_j(v))$. Hence $Z_i = Z_j$ if and only if

$$g_{ij}(x)\gamma_i(x)(w_i(v), w_i(v)) - \gamma_j(x)(w_j(v), w_j(v)) = (dg_{ij})_x(\varphi_x(v))w_i(v)$$

for every $v \in E_x$. Multiplying this identity by the frame $e_j(x)$ turns it into an identity in E_x , whose left-hand side becomes $\Gamma_{i,x}(v, v) - \Gamma_{j,x}(v, v)$ since $e_j g_{ij} = e_i$. Thus we obtain

$$\begin{aligned} \Gamma_{i,x}(v, v) - \Gamma_{j,x}(v, v) &= e_j(x)(dg_{ij})_x(\varphi_x(v))w_i(v) \\ &= e_i(x)g_{ij}(x)^{-1}(dg_{ij})_x(\varphi_x(v))w_i(v) \\ &= A_{ij,x}(\varphi_x(v))v \\ &= C_{ij,x}(v, v). \end{aligned}$$

Therefore, the vector fields Z_i and Z_j agree precisely when

$$\Gamma_{i,x}(v, v) - \Gamma_{j,x}(v, v) = C_{ij,x}(v, v) \quad (v \in E_x).$$

Since both sides are symmetric bilinear tensors, equality on all pairs (v, v) implies equality on all pairs by polarization. This proves the gluing criterion.

Step 3. Conclusion. Suppose that (2) holds. Since $\{C_{ij}\}$ represents the zero class, there exist sections Γ_i such that $\Gamma_i - \Gamma_j = C_{ij}$. Let γ_i be their coefficient expressions in the frames e_i , and define Z_i by (1) and (2). By Step 2, these vector fields glue together to a vector field Z on E , which is a quadratic φ -vector field by its local expression. This proves (1).

Conversely, suppose that (1) holds, and let Z be a quadratic φ -vector field. Write Z locally as in (1) and (2), and let Γ_i be the corresponding tensors. These local expressions agree on overlaps since they come from the same global vector field. By Step 2, we have $\Gamma_i - \Gamma_j = C_{ij}$. Hence $\{C_{ij}\}$ is a coboundary, so $\alpha_\varphi(E) = 0$, which proves (2). $\square$

Corollary 2.7. *If $\varphi: E \rightarrow T_X$ is a surjective bundle morphism and the symmetrized φ -Atiyah class $\alpha_\varphi(E)$ vanishes, then there exists a local dominating spray $s: U \rightarrow X$.*

Proof. By Theorem 2.6, E admits a quadratic φ -vector field. Its exponential φ -spray is a local spray with fibre derivative φ by Proposition 2.4. Since φ is surjective, this spray is dominating. $\square$

Remark 2.8. The vanishing of $\alpha_\varphi(E)$ is necessary and sufficient for the existence of a quadratic φ -vector field, but it is only a sufficient condition for the existence of a local spray with fibre derivative φ (not necessarily an exponential φ -spray).

Indeed, let $X = \mathbb{C}/\Lambda$ be an elliptic curve, let L be a line bundle on X with $\deg L \neq 0$, and set

$$E := T_X \oplus L, \quad \varphi: E \longrightarrow T_X, \quad \varphi(u, \ell) = u.$$

The translation trivialization $T_X \cong X \times \mathbb{C}$ defines a global dominating spray

$$s: E \longrightarrow X, \quad s(x, u, \ell) = x + u \pmod{\Lambda},$$

whose fibre derivative is φ . We show that $\alpha_\varphi(E) \neq 0$.

Choose a global translation-invariant frame of T_X and local frames e_i^L of L , with transition functions h_{ij} defined by $e_i^L = e_j^L h_{ij}$. The corresponding transition matrices of E and the cocycle representing its Atiyah class are

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & h_{ij} \end{pmatrix}, \quad A_{ij} = g_{ij}^{-1} dg_{ij} = \begin{pmatrix} 0 & 0 \\ 0 & A_{ij}^L \end{pmatrix}, \quad A_{ij}^L := h_{ij}^{-1} dh_{ij}.$$

Here $\{A_{ij}^L\}$ represents $a(L) \in H^1(X, \Omega_X^1)$, where we use $\text{End } L \cong \mathcal{O}_X$. In particular, for $\xi \in T_x X$ and $(u, \ell) \in E_x$,

$$A_{ij,x}(\xi)(u, \ell) = (0, A_{ij,x}^L(\xi)\ell).$$

As in the proof of Theorem 2.6, the class $\alpha_\varphi(E)$ is represented by

$$C_{ij}(a, b) = \frac{1}{2}(A_{ij}(\varphi(a))b + A_{ij}(\varphi(b))a).$$

For $u \in T_x X$ and $\ell \in L_x$, we have $\varphi(u, 0) = u$ and $\varphi(0, \ell) = 0$. Consequently,

$$\begin{aligned} C_{ij}((u, 0), (0, \ell)) &= \frac{1}{2}(A_{ij}(u)(0, \ell) + A_{ij}(0)(u, 0)) \\ &= \left(0, \frac{1}{2}A_{ij}^L(u)\ell\right). \end{aligned}$$

To interpret this identity in cohomology, we use the direct sum decomposition $E = T_X \oplus L$ to write

$$S^2 E^\vee \cong S^2 T_X^\vee \oplus (T_X^\vee \otimes L^\vee) \oplus (L^\vee)^{\otimes 2}.$$

Tensoring with $E = T_X \oplus L$, we obtain

$$\begin{aligned} S^2 E^\vee \otimes E &\cong (S^2 T_X^\vee \otimes T_X) \oplus (S^2 T_X^\vee \otimes L) \\ &\quad \oplus (T_X^\vee \otimes L^\vee \otimes T_X) \oplus (T_X^\vee \otimes L^\vee \otimes L) \\ &\quad \oplus ((L^\vee)^{\otimes 2} \otimes T_X) \oplus ((L^\vee)^{\otimes 2} \otimes L). \end{aligned}$$

The fourth summand consists of the mixed terms with values in L and is canonically isomorphic to Ω_X^1 since $L^\vee \otimes L \cong \mathcal{O}_X$. Under this identification, a 1-form ω corresponds to the bilinear map $(u, \ell) \mapsto \omega(u)\ell$. The preceding calculation shows that the component of C_{ij} in this summand is $\frac{1}{2}A_{ij}^L$. Thus, in the induced direct sum decomposition of $H^1(X, S^2 E^\vee \otimes E)$, the component of $\alpha_\varphi(E)$ in

$$H^1(X, T_X^\vee \otimes L^\vee \otimes L) \cong H^1(X, \Omega_X^1)$$

is $(1/2)a(L)$. Since $a(L) \neq 0$ by $\deg L \neq 0$, it follows that $\alpha_\varphi(E) \neq 0$. By Theorem 2.6, E admits no quadratic φ -vector field, and hence no exponential φ -spray, although it admits the global dominating spray defined above.

2.2. Examples admitting no local dominating spray. Theorem 4.1 shows that every weakly pseudoconvex manifold admitting a positive line bundle has a local dominating spray. However, a compact Kähler manifold without a positive line bundle need not admit a local dominating spray. Thus the positivity hypothesis cannot be omitted in general.

Example 2.9 (Compact Kähler manifolds without local dominating sprays). Let $\mathbb{T}$ be a complex torus of algebraic dimension zero, and let $\pi: X := \text{Bl}_{t_0}(\mathbb{T}) \rightarrow \mathbb{T}$ be the blow-up at a point $t_0 \in \mathbb{T}$. Then X admits no local dominating spray.

Proof. After a translation of $\mathbb{T}$, we may assume that $t_0 = 0$. Write $\mathbb{T} = \mathbb{C}^n/A$; note that $n \geq 2$, since complex tori of dimension one are projective. Let $\text{Exc}(\pi) := \pi^{-1}(0)$ be the exceptional divisor. Suppose that a vector bundle $p: E \rightarrow X$ carries a local spray $s: U \rightarrow X$ whose fibre derivative $\varphi: E \rightarrow T_X$ is surjective. Set $r := \text{rank } E \geq n$ and $X^\circ := X \setminus \text{Exc}(\pi)$.

We outline the argument. We consider a *displacement* δ defined by the dominating spray s . The coefficients of the Taylor expansion of δ along the fibres impose a rank bound along the fibres of an algebraic reduction of $\mathbf{P}(E)$ (see Step 4). On the other hand, we construct a hypersurface S whose image $s(S)$ is contained in $\text{Exc}(\pi)$ (see Step 5). On S , the displacement $\delta(v)$ depends only on the base point $p(v)$, and its differential has rank n in the same directions, which gives a contradiction (see Step 8).

Step 1. The displacement and its Taylor coefficients. Via the natural quotient map $\mathbb{C}^n \rightarrow \mathbb{T} = \mathbb{C}^n/\Lambda$, we identify a small neighbourhood of the origin in $\mathbb{T}$ with a neighbourhood of the origin in $\mathbb{C}^n$, and we regard the points of this neighbourhood as vectors. Since $s(0_x) = x$, we may shrink U to a fibrewise ball bundle such that $\pi(s(v)) - \pi(p(v))$ lies in this origin chart for every $v \in U$. We define the displacement

$$\delta(v) := \pi(s(v)) - \pi(p(v)) \in \mathbb{C}^n \quad (v \in U \subset E),$$

where the difference is expressed in the fixed origin coordinates. Since δ vanishes on the zero section, its fibrewise Taylor expansion is

$$(8) \quad \delta(v) = \sum_{k \geq 1} \delta_k(v), \quad \delta_k(v) := \frac{1}{k!} \left. \frac{d^k}{dt^k} \right|_{t=0} \delta(tv).$$

The derivatives are taken componentwise in $\mathbb{C}^n$. For every $v \in E_x$, the formula defining $\delta_k(v)$ makes sense since $tv \in U$ for sufficiently small t . Each δ_k is a homogeneous polynomial of degree k on every fibre and corresponds to a global section of $S^k E^\vee \otimes (X \times \mathbb{C}^n)$. The series in (8) converges locally uniformly on U .

Step 2. A coordinate map near the exceptional divisor. Choose $\rho > 0$ so small that $B_{2\rho} \subset \mathbb{C}^n$ lies in the origin chart fixed above, and regard B_ρ as an open subset of $\mathbb{T}$. Define

$$Q_0: N_0 := \pi^{-1}(B_\rho) \xrightarrow{\pi} B_\rho \subset \mathbb{C}^n.$$

Then Q_0 vanishes on $\text{Exc}(\pi)$ and has differential of rank n on $N_0 \setminus \text{Exc}(\pi)$. For $x, y \in N_0$, the torus difference $\pi(y) - \pi(x)$ is represented in the origin chart by $Q_0(y) - Q_0(x)$.

Step 3. Fixing the algebraic parameter. Let $q: \mathbf{P}(E) \rightarrow X$ be the bundle of lines in E , and let $\nu: E \setminus 0_X \rightarrow \mathbf{P}(E)$ be the natural projection, so that $q \circ \nu = p$. Choose an algebraic reduction $\alpha: \mathbf{P}(E) \dashrightarrow Z$, where Z is a smooth projective variety. Every meromorphic function on $\mathbf{P}(E)$ is the pullback under α of a rational function on Z (see [22]). The relevant maps are shown in the following diagram:

$$\begin{array}{ccccc} \mathbf{P}(E) & \overset{\alpha}{\dashrightarrow} & Z & & \\ \downarrow q & \swarrow \nu & & & \\ X & \xleftarrow{p} & E & \xleftarrow{\quad} & E \setminus 0_X \\ \downarrow \pi & \swarrow s & \uparrow & & \\ \mathbb{T} & & U & & \end{array}$$

We show that $(\pi \circ q, \alpha): \mathbf{P}(E) \dashrightarrow \mathbb{T} \times Z$ is a dominant meromorphic map. Let $G \subset \mathbb{T} \times Z$ be the closure of its image, obtained from the closure of its graph. It is an irreducible

compact analytic subset surjecting onto both factors. Over a dense Zariski open subset $V \subset \mathbb{T}$, the projection $G \rightarrow \mathbb{T}$ is flat, and its fibres $G_t \subset Z$ have a common Hilbert polynomial Q . They therefore define a meromorphic map $\mathbb{T} \dashrightarrow \text{Hilb}^Q(Z)$. Since the target is projective and $a(\mathbb{T}) = 0$, this map is constant. Thus $G_t = Z'$ for every $t \in V$, for some fixed closed algebraic subset $Z' \subset Z$. Since G is irreducible, $G \cap (V \times Z)$ is dense in G . Taking closures, we obtain

$$G = \overline{G \cap (V \times Z)} = \overline{V \times Z'} = \mathbb{T} \times Z.$$

Hence the differential of $(\pi \circ q, \alpha)$ is surjective at a general point of $\mathbf{P}(E)$. Since $d\pi$ is an isomorphism on X° , we obtain

$$(9) \quad dq_{[\ell]}(\ker d\alpha_{[\ell]}) = T_{q([\ell])}X$$

at a general point $[\ell] \in \mathbf{P}(E)$. Thus the base point in X can move in all n tangent directions while the value of α remains fixed.

Step 4. The rank bound from the Taylor coefficients. For $v \in E_x$, differentiating the displacement at 0_x in the fibre direction gives

$$\delta_1(v) = \left. \frac{d}{dt} \right|_{t=0} (\pi(s(tv)) - \pi(x)) = d\pi_x(\varphi_x(v)).$$

Here the last term is identified with a vector in $\mathbb{C}^n$ via translation from $\pi(x)$ to the origin and the fixed origin coordinates. Since φ_x is surjective and $d\pi_x$ is an isomorphism for $x \in X^\circ$, the linear map $\delta_1|_{E_x}: E_x \rightarrow \mathbb{C}^n$ is surjective for $x \in X^\circ$. Hence its first component $\gamma := \delta_1^1$ is a nonzero section of $E^\vee$.

Work on the dense open subset of $\mathbf{P}(E)$ where α is a holomorphic submersion, the base point lies in X° , and γ is nonzero. For $k \geq 1$, each component of δ_k/γ^k is homogeneous of degree zero and defines a meromorphic function on $\mathbf{P}(E)$. Each component can be written as the pullback under α of a rational function on Z . Thus its differential annihilates $\ker d\alpha$, first where the rational function is holomorphic and then on the whole open subset by holomorphic continuation.

Pulling back by ν and differentiating the ratio, we obtain

$$\begin{aligned} (d\delta_k)_\ell(\eta) &= d \left[\gamma^k \left(\frac{\delta_k}{\gamma^k} \circ \nu \right) \right]_\ell(\eta) \\ &= k \frac{\delta_k(\ell)}{\gamma(\ell)} d\gamma_\ell(\eta) \quad (\eta \in \ker d(\alpha \circ \nu)_\ell). \end{aligned}$$

Here we regard δ_k as a holomorphic map $E \rightarrow \mathbb{C}^n$. Thus all $d\delta_k$ vanish on $\ker d\gamma \cap \ker d(\alpha \circ \nu)$. Differentiating (8) term by term, we deduce that

$$\text{rank} \left((d\delta)_\ell|_{\ker d(\alpha \circ \nu)_\ell} \right) \leq 1$$

since $d\gamma$ is scalar-valued.

Step 5. A hypersurface mapped into the exceptional divisor. Fix $x_0 \in \text{Exc}(\pi)$, choose local coordinates $z = (z_1, \dots, z_n)$ centred at x_0 with $\text{Exc}(\pi) = \{z_1 = 0\}$, and trivialize E with fibre coordinates $w = (w_1, \dots, w_r)$. By surjectivity of φ_{x_0} , we may assume that

$\partial s^1 / \partial w_1(0, 0) \neq 0$. Shrink the product neighbourhood in E so that its image under s lies in this coordinate chart in N_0 and this derivative is nonzero there.

For $b = (b_2, \dots, b_r)$ near zero, the implicit function theorem shows that the equation

$$s^1(z, t(1, b_2, \dots, b_r)) = 0$$

has a unique small solution $t = t(z, b)$, which depends holomorphically on (z, b) . Since $s^1(z, 0) = z_1$, the solution $t(z, b)$ vanishes exactly when $z_1 = 0$. For $z_1 \neq 0$, we consider the holomorphic map

$$(z, b) \mapsto (z, t(z, b)(1, b_2, \dots, b_r)) \in E$$

and define S to be its image. Thus S is a local hypersurface in E satisfying $s(S) \subset \text{Exc}(\pi)$ and $p(S) \subset X^\circ$. In the projective chart $w_1 \neq 0$, the restriction $\nu|_S$ is given by $(z, b) \mapsto (z, b)$. Hence $\nu(S)$ is open in $\mathbf{P}(E)$, and $\nu|_S: S \rightarrow \nu(S)$ is biholomorphic.

The solutions $t(z, 0)$ tend to zero as $z \rightarrow x_0$. Thus we may choose a small simply connected neighbourhood $V \subset N_0 \setminus \text{Exc}(\pi)$ and a fixed fibre polydisc D_ε^r such that

$$\mathcal{A} := V \times D_\varepsilon^r \subset U, \quad s(\mathcal{A}) \subset N_0, \quad S \cap \mathcal{A} \neq \emptyset.$$

Replace S by a nonempty open set contained in $\mathcal{A}$. The key point is that the same connected product neighbourhood $\mathcal{A}$ contains both the zero section over V and this nonempty patch of S . Although this choice of $\mathcal{A}$ is not essential in the torus blow-up case, we make it here in preparation for the next example.

Step 6. The displacement on the hypersurface. Since both $p(v)$ and $s(v)$ lie in N_0 for $v \in \mathcal{A}$, the coordinate identity from Step 2 gives $\delta|_{\mathcal{A}} = Q_0 \circ s - Q_0 \circ p$. On S , the first term vanishes, so

$$\delta|_S = -Q_0 \circ p|_S.$$

Thus the displacement on S depends only on the base point $p(v)$.

Step 7. Extending the rank bound by analytic continuation. Put $\mathcal{A}^\times := \mathcal{A} \setminus 0_X$. In the chosen trivialization, $\mathcal{A}^\times = V \times (D_\varepsilon^r \setminus \{0\})$, so $\mathcal{A}^\times$ is connected. By Step 4, we have $\text{rank} \left(d\delta|_{\ker d(\alpha \circ \nu)} \right) \leq 1$ on a nonempty open subset of $\mathcal{A}^\times$. By analytic continuation, this inequality holds wherever $\alpha \circ \nu$ is a holomorphic submersion.

Step 8. The differential has rank n on S . Since $\nu(S)$ is open in $\mathbf{P}(E)$, we can choose $\ell \in S$ such that α is a holomorphic submersion at $[\ell] := \nu(\ell)$ and (9) holds there. The map $\nu|_S$ is locally biholomorphic and identifies $T_\ell S \cap \ker d(\alpha \circ \nu)_\ell$ with $\ker d\alpha|_{[\ell]}$. Since $p = q \circ \nu$, we obtain

$$dp_\ell(T_\ell S \cap \ker d(\alpha \circ \nu)_\ell) = T_{p(\ell)}X.$$

By Step 6, we have $\delta = -Q_0 \circ p$ on S . Since $p(\ell) \in V \subset N_0 \setminus \text{Exc}(\pi)$ and $(dQ_0)_{p(\ell)}$ has rank n , the preceding surjectivity yields

$$\text{rank} \left((d\delta)_\ell|_{T_\ell S \cap \ker d(\alpha \circ \nu)_\ell} \right) = n.$$

This contradicts the rank bound in Step 7, since $n \geq 2$. Thus X admits no local dominating spray. $\square$

Example 2.10 (Kummer surfaces without local dominating sprays). Let $\mathbb{T} = \mathbb{C}^2/\Lambda$ be a two-dimensional complex torus of algebraic dimension zero, and let X be its Kummer surface. Then X admits no local dominating spray.

Proof. We note that $\mathbb{T}$ contains no curves, and hence no nontrivial subvarieties. Indeed, let $C \subset \mathbb{T}$ be an irreducible curve. Since $K_{\mathbb{T}}$ is trivial, the adjunction formula gives $C^2 = 2p_a(C) - 2$. As $\mathbb{T}$ contains no rational curves, we have $p_a(C) \geq 1$, and hence $C^2 \geq 0$. If $C^2 > 0$, then the line bundle $\mathcal{O}_{\mathbb{T}}(C)$ is big, and hence $\mathbb{T}$ is Moishezon. If $C^2 = 0$, then $p_a(C) = 1$ and C is a smooth elliptic curve, which is a translate of a one-dimensional subtorus $\mathbb{T}'$; then the quotient map $\mathbb{T} \rightarrow \mathbb{T}/\mathbb{T}'$ onto an elliptic curve yields nonconstant meromorphic functions on $\mathbb{T}$. In either case, $a(\mathbb{T}) \geq 1$, a contradiction.

Let $\mathbb{T}[2] := (1/2)\Lambda/\Lambda$ be the set of the sixteen two-torsion points, and let $\pi: \tilde{\mathbb{T}} := \text{Bl}_{\mathbb{T}[2]} \mathbb{T} \rightarrow \mathbb{T}$ be the blow-up of $\mathbb{T}$ at these points. The involution $y \mapsto -y$ lifts to an involution $\tilde{\iota}$ of $\tilde{\mathbb{T}}$. The quotient $X = \tilde{\mathbb{T}}/\tilde{\iota}$ is a K3 surface, and the quotient map $\tau: \tilde{\mathbb{T}} \rightarrow X$ is a double cover branched along the curves $\Gamma_b := \tau(\pi^{-1}(b))$. Set $\Gamma := \bigcup_{b \in \mathbb{T}[2]} \Gamma_b$ and $X^\circ := X \setminus \Gamma$.

Fix $a \in \mathbb{T}[2]$. Suppose that a vector bundle $p: E \rightarrow X$ of rank r carries a local spray $s: U \rightarrow X$ whose fibre derivative $\varphi: E \rightarrow T_X$ is surjective at every point of Γ_a .

As in Example 2.9, we use a displacement whose Taylor coefficients impose a rank bound along the fibres of an algebraic reduction of $\mathbf{P}(E)$. We then construct a hypersurface S that s maps into Γ_a . Passing to a branched double cover makes the displacement holomorphic over S , where its differential has rank two in the same directions; this gives a contradiction.

Step 1. The squared displacement. Via the natural quotient map $\mathbb{C}^2 \rightarrow \mathbb{T}$, we identify a small neighbourhood of the origin in $\mathbb{T}$ with a neighbourhood of the origin in $\mathbb{C}^2$, and we use this identification to regard small torus differences as vectors.

The maps π and τ induce a natural unramified double cover $\tau^\circ: \mathbb{T}^\circ := \mathbb{T} \setminus \mathbb{T}[2] \rightarrow X^\circ$, whose two sheets are interchanged by $y \mapsto -y$.

Take a fibrewise ball bundle $U^\circ \subset U \cap p^{-1}(X^\circ)$ such that $s(U^\circ) \subset X^\circ$. For $x \in X^\circ$, we lift $s|_{U_x^\circ}$ to a holomorphic map $\tilde{s}_x: U_x^\circ \rightarrow \mathbb{T}^\circ$, choosing one point of the preimage of x as its initial value. After a further fibrewise shrinking, we define a holomorphic map

$$\delta: U^\circ \rightarrow S^2\mathbb{C}^2 \quad \text{by } \delta(v) := (\tilde{s}_x(v) - \tilde{s}_x(0_x))^{\otimes 2} \in S^2\mathbb{C}^2, \quad (v \in U_x^\circ).$$

Here the difference is expressed in the fixed coordinates near the origin, and $u^{\otimes 2} = (u_1^2, u_1u_2, u_2^2)$ for $u = (u_1, u_2)$. This δ is well defined and holomorphic. Indeed, changing the initial value changes the signs of both terms but leaves δ unchanged. Moreover, the lifts vary holomorphically over small simply connected neighbourhoods in X° , so these expressions define a holomorphic map $\delta: U^\circ \rightarrow S^2\mathbb{C}^2$.

Since the difference vanishes on the zero section, its square has the fibrewise Taylor expansion

$$(10) \quad \delta(v) = \sum_{k \geq 2} \delta_k(v), \quad \delta_k(v) := \frac{1}{k!} \left. \frac{d^k}{dt^k} \right|_{t=0} \delta(tv).$$

Each δ_k is a homogeneous polynomial of degree k on every fibre and corresponds to a section of $S^k E^\vee \otimes (X^\circ \times S^2 \mathbb{C}^2)$ over X° . The series in (10), together with its first derivatives, converges locally uniformly on U° .

Step 2. Meromorphic Taylor coefficients. In this step, we prove that $\delta_k z_1^k$ extends holomorphically across Γ_b , where z_1 is a local defining function of Γ_b , and hence that δ_k is a meromorphic section of $S^k E^\vee \otimes (X \times S^2 \mathbb{C}^2)$. To this end, we first describe the domain of δ precisely.

Choose $\rho > 0$ so small that $B_{2\rho} \subset \mathbb{C}^2$ lies in the origin chart fixed above and the neighbourhoods $b + B_\rho \subset \mathbb{T}$, $b \in \mathbb{T}[2]$, are pairwise disjoint. For each b , we consider the coordinate map

$$u_b: b + B_\rho \rightarrow B_\rho \quad \text{defined by } b + u \mapsto u$$

and put

$$\tilde{N}_b := \pi^{-1}(b + B_\rho), \quad N_b := \tau(\tilde{N}_b).$$

The restriction of the natural double cover gives $b + B_\rho \setminus \{b\} \rightarrow N_b \setminus \Gamma_b$. For two points of $b + B_\rho$, their torus difference is represented by the difference of their u_b -coordinates.

Near a point of Γ_b , choose coordinates $z = (z_1, z_2)$ with $\Gamma_b = \{z_1 = 0\}$ and trivialize E with fibre coordinates $w \in \mathbb{C}^r$. Since $s(z, 0) = z$, we have $|s^1(z, w) - z_1| \leq C|w|$ for some uniform constant C . Fix $c > 0$ so small that, for $z_1 \neq 0$, the full fibre ball $D_z := \{w : |w| < c|z_1|\}$ satisfies

$$|s^1(z, w)| \geq \frac{1}{2}|z_1| > 0 \quad (w \in D_z).$$

Thus $s(z, D_z)$ avoids Γ_b , and the map $w \mapsto s(z, w)$ on D_z lifts to $b + B_\rho \setminus \{b\}$ via the unramified cover $b + B_\rho \setminus \{b\} \rightarrow N_b \setminus \Gamma_b$; we denote this lift by $\tilde{s}_z$. Near $w = 0$, the displacement from Step 1 is given by

$$\delta(z, w) = (u_b(\tilde{s}_z(w)) - u_b(\tilde{s}_z(0)))^{\otimes 2}.$$

The right-hand side is defined on all of D_z , independently of the choice of lift, and each component is bounded by $4\rho^2$, uniformly in z .

Consider a component $\delta_k^{ij}(z, w) = \sum_{|\mu|=k} a_\mu^{ij}(z) w^\mu$ of δ_k , where $1 \leq i \leq j \leq 2$ and μ is a multi-index. Since the closed fibre polydisc of radius $c|z_1|/(2\sqrt{r})$ lies in D_z , Cauchy's estimates for the coefficients give

$$|a_\mu^{ij}(z)| \leq 4\rho^2 \left(\frac{2\sqrt{r}}{c|z_1|} \right)^k \quad (|\mu| = k).$$

Consequently, by the Riemann extension theorem, $z_1^k a_\mu^{ij}$ extends holomorphically across Γ_b . Thus each δ_k extends meromorphically across Γ_b with poles of order at most k . Note that only the individual Taylor coefficients are extended.

For later use, we introduce the squared coordinate map

$$Q_b: N_b \rightarrow S^2 \mathbb{C}^2 \quad \text{defined by } Q_b(\tau(\tilde{y})) := (u_b(\pi(\tilde{y})))^{\otimes 2}.$$

It is well defined and holomorphic since $b = -b$ and $u_b(-y) = -u_b(y)$. In a blow-up chart, write $u_b \circ \pi = (t, t\zeta)$. The quotient coordinates are then $z_1 = t^2$ and $z_2 = \zeta$, which

gives $Q_b(z_1, z_2) = (z_1, z_1 z_2, z_1 z_2^2)$. This expression shows that Q_b vanishes on Γ_b and has differential of rank two on $N_b \setminus \Gamma_b$.

Step 3. Fixing the algebraic parameter. Let $q: \mathbf{P}(E) \rightarrow X$ be the bundle of lines in E , and let $\nu: E \setminus 0_X \rightarrow \mathbf{P}(E)$ be the natural projection, so that $q \circ \nu = p$. Choose an algebraic reduction $\alpha: \mathbf{P}(E) \dashrightarrow Z$ such that Z is a smooth projective variety.

We show that $(q, \alpha): \mathbf{P}(E) \dashrightarrow X \times Z$ is dominant, as in Step 3 of Example 2.9. Let $G \subset X \times Z$ be the closure of the image of (q, α) . It is an irreducible compact analytic subset surjecting onto both factors. Suppose that $G \neq X \times Z$. Then the general fibres of $G \rightarrow Z$ are proper analytic subsets of X . Since $\mathbb{T}$ contains no curves, every curve in X is contained in Γ ; indeed, the image under π of an irreducible curve in $\tilde{\mathbb{T}}$ not contained in the exceptional set would be a curve in $\mathbb{T}$. If the general fibres are finite, then $G \rightarrow Z$ is generically finite, which makes G and its image X Moishezon; this contradicts $a(X) = a(\tilde{\mathbb{T}}) = a(\mathbb{T}) = 0$. Otherwise, the general fibres are contained in Γ , and hence $G \subset \Gamma \times Z$, which contradicts the surjectivity of $G \rightarrow X$. Thus (q, α) is dominant. Hence its differential is surjective at a general point, and therefore

$$(11) \quad dq_{[\ell]}(\ker d\alpha_{[\ell]}) = T_{q([\ell])}X$$

at a general point $[\ell] \in \mathbf{P}(E)$.

Step 4. The rank bound from the Taylor coefficients. Since δ is the square of the displacement, its degree-two term is

$$\delta_2(v) = \left(\frac{d}{dt} \Big|_{t=0} (\tilde{s}_x(tv) - \tilde{s}_x(0_x)) \right)^{\otimes 2} \quad (v \in E_x),$$

where the derivative is computed in the fixed origin chart. The assumption implies that $(d/dt)|_{t=0} \tilde{s}_x(tv)$ is surjective on a neighbourhood of Γ_a . Hence $\gamma := \delta_2^{11}$ is a nonzero meromorphic section of $S^2 E^\vee$.

Work on the dense open subset of $\mathbf{P}(E)$ where α is a holomorphic submersion, the base point lies in X° , and γ is nonzero. For $k \geq 2$ and $1 \leq i \leq j \leq 2$, the ratio $(\delta_k^{ij})^2/\gamma^k$ is homogeneous of degree zero and defines a meromorphic function on $\mathbf{P}(E)$; hence it can be written as the pullback under α of a rational function on Z . This shows that

$$(d\delta_k^{ij})_\ell(\eta) = \frac{k\delta_k^{ij}(\ell)}{2\gamma(\ell)} d\gamma_\ell(\eta) \quad (\eta \in \ker d(\alpha \circ \nu)_\ell).$$

Thus all $d\delta_k$ vanish on $\ker d\gamma \cap \ker d(\alpha \circ \nu)$. Differentiating (10) term by term gives the same conclusion for $d\delta$ on U° . Therefore, since $d\gamma$ is scalar-valued, we obtain

$$(12) \quad \text{rank} \left((d\delta)_\ell \Big|_{\ker d(\alpha \circ \nu)_\ell} \right) \leq 1.$$

Step 5. A hypersurface mapped into Γ_a . Fix $x_0 \in \Gamma_a$, choose coordinates $z = (z_1, z_2)$ centred at x_0 with $\Gamma_a = \{z_1 = 0\}$, and trivialize E with fibre coordinates $w = (w_1, \dots, w_r)$. By the surjectivity of φ_{x_0} , we may choose the frame so that $\partial s^1/\partial w_1(0, 0) \neq 0$. Shrink the product neighbourhood so that its image under s lies in this coordinate chart in N_a and this derivative is nonzero throughout.

Applying the same implicit-function construction as in Step 5 of the preceding example, we obtain a nonempty patch $S \subset s^{-1}(\Gamma_a)$ with $p(S) \subset X^\circ$ such that $\nu|_S$ is locally

biholomorphic. As in that construction, choose a small simply connected neighbourhood $V \subset N_a \setminus \Gamma$ and a fixed fibre polydisc D_ε^r such that

$$\mathcal{A} := V \times D_\varepsilon^r \subset U, \quad s(\mathcal{A}) \subset N_a, \quad S \cap \mathcal{A} \neq \emptyset.$$

Replace S by a nonempty patch contained in $\mathcal{A}$.

The point is that the displacement δ , which is defined near the zero section over V , cannot be defined along S . Indeed, the lift used to define it does not extend across S : in local coordinates, such an extension would require a holomorphic square root of s^1 , which vanishes to first order along S . To overcome this difficulty, in the next step we extend the displacement on a branched cover.

Step 6. Extending the displacement on a branched cover. Consider the fibre product

$$\widehat{\mathcal{A}} := \{(v, \widetilde{y}) \in \mathcal{A} \times \widetilde{N}_a : s(v) = \tau(\widetilde{y})\}, \quad \text{pr}(v, \widetilde{y}) := v,$$

which fits into the Cartesian diagram

$$\begin{array}{ccc} \widehat{\mathcal{A}} & \xrightarrow{(v, \widetilde{y}) \mapsto \widetilde{y}} & \widetilde{N}_a \\ \text{pr} \downarrow & (t, \zeta) \mapsto (t^2, \zeta) & \downarrow \tau \\ \mathcal{A} & \xrightarrow{s} & N_a \end{array}$$

In these coordinates, the subvariety $\widehat{\mathcal{A}}$ is defined by $t^2 - s^1(z, w) = 0$, which shows, since $\partial s^1 / \partial w_1 \neq 0$, that S is a smooth hypersurface.

The projection pr is a double cover, ramified precisely along $t = 0$, so its ramification locus is

$$R := \{(v, \widetilde{y}) \in \widehat{\mathcal{A}} : \widetilde{y} \in \pi^{-1}(a)\}.$$

On R , we have $s^1(z, w) = 0$ and $\zeta = s^2(z, w)$. Hence each $v \in s^{-1}(\Gamma_a) \cap \mathcal{A}$ determines a unique point of R , and the resulting inverse is holomorphic. Consequently, $\text{pr}|_R : R \rightarrow s^{-1}(\Gamma_a) \cap \mathcal{A}$ is an isomorphism.

Since $V \subset X^\circ$ is simply connected, we obtain a lift of the points of V to $a + B_\rho \setminus \{a\}$, and we denote its coordinate representation by $u : V \rightarrow B_\rho \setminus \{0\}$. By the construction of Q_a (see Step 2), we have $u(x)^{\otimes 2} = Q_a(x)$. Define

$$\widehat{\delta}(v, \widetilde{y}) := (u_a(\pi(\widetilde{y})) - u(p(v)))^{\otimes 2} \quad ((v, \widetilde{y}) \in \widehat{\mathcal{A}}).$$

This map is holomorphic on all of $\widehat{\mathcal{A}}$, including R , although δ is not defined along S .

Near the lift of the zero section on which $u_a(\pi(\widetilde{y})) = u(x)$ at $v = 0_x$, the target lift coincides with the fibrewise lift from Step 1 with initial coordinate $u(x)$. Hence, by the definition of δ , we have $\widehat{\delta} = \delta \circ \text{pr}$ on a neighbourhood of this lifted zero section. We use this equality only on this initial sheet.

On R , the lifted target satisfies $\pi(\widetilde{y}) = a$, so its coordinate vanishes. Hence

$$(13) \quad \widehat{\delta}(v, \widetilde{y}) = (-u(p(v)))^{\otimes 2} = Q_a(p(v)), \quad (v, \widetilde{y}) \in R.$$

Thus the displacement on R depends only on the base point $p(v)$.

Step 7. Extending the rank bound by analytic continuation.

Removing the inverse image of the zero section, we put $\widehat{\mathcal{A}}^\times := \widehat{\mathcal{A}} \setminus \text{pr}^{-1}(0_X)$. Note that $\widehat{\mathcal{A}}^\times$ is still connected. On a nonempty open subset of the initial sheet from Step 6, the map pr is locally biholomorphic and $\widehat{\delta} = \delta \circ \text{pr}$. The rank bound (12) gives

$$(14) \quad \text{rank} \left(d\widehat{\delta} \big|_{\ker d(\alpha \circ \nu \circ \text{pr})} \right) \leq 1$$

on the locus where $\alpha \circ \nu \circ \text{pr}$ is a holomorphic submersion.

Step 8. The differential has rank two on R . Since $\nu(S)$ is open in $\mathbf{P}(E)$, we can choose $\ell \in S$ such that α is a holomorphic submersion at $[\ell] := \nu(\ell)$ and (11) holds there. Let $\widehat{\ell} \in R$ be the unique point over ℓ . The restriction of $\nu \circ \text{pr}$ to R is locally biholomorphic at $\widehat{\ell}$ and identifies $T_{\widehat{\ell}}R \cap \ker d(\alpha \circ \nu \circ \text{pr})_{\widehat{\ell}}$ with $\ker d\alpha_{[\ell]}$. Since $p = q \circ \nu$, we obtain

$$d(p \circ \text{pr})_{\widehat{\ell}}(T_{\widehat{\ell}}R \cap \ker d(\alpha \circ \nu \circ \text{pr})_{\widehat{\ell}}) = T_{p(\ell)}X.$$

On R , (13) gives $\widehat{\delta} = Q_a \circ p \circ \text{pr}$. Since $p(\ell) \in V \subset N_a \setminus \Gamma_a$ and $(dQ_a)_{p(\ell)}$ has rank two, the preceding surjectivity yields

$$\text{rank} \left(d\widehat{\delta} \big|_{T_{\widehat{\ell}}R \cap \ker d(\alpha \circ \nu \circ \text{pr})_{\widehat{\ell}}} \right) = 2.$$

This contradicts (14). Thus φ fails to be surjective at some point of Γ_a . Since $a \in \mathbb{T}[2]$ was arbitrary, this holds for every exceptional curve. In particular, X admits no local dominating spray. $\square$

Remark 2.11 (Oka K3 surfaces that are not elliptic, and non-openness of ellipticity). Example 2.10, combined with the recent work of Xie and Zhao [23], has the following consequences. Note that a two-dimensional complex torus $\mathbb{T}$ has algebraic dimension zero if its Néron–Severi group vanishes. Indeed, if $a(\mathbb{T}) \geq 1$, then a nonconstant meromorphic function on $\mathbb{T}$ has a nonzero polar divisor, whose class in $\text{NS}(\mathbb{T})$ is nonzero since $\mathbb{T}$ is Kähler.

(1) *Oka K3 surfaces that are not elliptic.* By [23, Theorem E (2)], there exist Oka Kummer surfaces whose Néron–Severi lattice coincides with the Kummer lattice. Since the Picard number of the Kummer surface of $\mathbb{T}$ equals $\rho(\mathbb{T}) + 16$, the underlying torus $\mathbb{T}$ of such a surface has Picard number zero. Since $H^2(\mathbb{T}, \mathbb{Z})$ is torsion-free, $\text{NS}(\mathbb{T}) = 0$, and hence $\mathbb{T}$ has algebraic dimension zero. By Example 2.10, these Oka K3 surfaces are not elliptic.

(2) *Ellipticity is not open in families.* The Fermat quartic $X_0 \subset \mathbb{P}^3$ is a Kummer surface by a theorem of Mizukami (see [13, Appendix, Theorem A.1]) and it is Oka by [23, Corollary 4.3]. Since X_0 is projective, it is elliptic by [10] (or by Theorem 1.1). Let $\mathbb{T}_0$ be a two-dimensional complex torus whose Kummer surface is X_0 . Complex tori $\mathbb{C}^2/\Lambda$ are parametrized by an open subset of the space of lattices, and those of positive algebraic dimension, that is, those containing a curve, form a countable union of proper analytic subsets of the parameter space. Hence there is a holomorphic family $\{\mathbb{T}_t\}_{t \in \mathbb{D}}$ of two-dimensional complex tori over the unit disc $\mathbb{D}$, with the given central fibre $\mathbb{T}_0$, such that $\mathbb{T}_t$ has algebraic dimension zero for every t outside a countable subset of $\mathbb{D}$. Performing the Kummer construction in this family (blowing up the sixteen sections of two-torsion points and taking the quotient by the fibrewise involution), we obtain a holomorphic family $\{X_t\}_{t \in \mathbb{D}}$ of Kummer surfaces, that is, a proper holomorphic submersion with fibres X_t .

Its central fibre X_0 is elliptic, whereas X_t is not elliptic for every t outside a countable subset of $\mathbb{D}$ by Example 2.10. Therefore ellipticity is not an open condition in holomorphic families of compact complex manifolds.

Remark 2.12 (Ellipticity is neither open nor closed in families of Oka manifolds). Example 2.9 shows that ellipticity is neither an open nor a closed condition in holomorphic families of compact Oka manifolds. For comparison, Forstnerič and Lárusson showed that the Oka property itself is not closed in families [9]; in our families, every fibre is Oka.

There is a holomorphic family $\{\mathbb{T}_b\}_{b \in B}$ of two-dimensional complex tori over a connected complex manifold B such that both the fibres that are projective and the fibres with vanishing Néron–Severi group occur over dense subsets of B . Indeed, take the family of all tori $\mathbb{C}^2/(I_2, b)\mathbb{Z}^4$ with $\det(\operatorname{Im} b) > 0$; its period map identifies B with an open subset of the period domain $Q = \{[\sigma] : \sigma^2 = 0, \sigma \cdot \bar{\sigma} > 0\}$ in $\mathbf{P}(H^2(\mathbb{T}, \mathbb{C}))$, where $\mathbb{T}$ is a fixed fibre (see the proof of [19, Theorem II]), and $\operatorname{NS}(\mathbb{T}_b)$ consists of the integral classes orthogonal to σ_b . For each nonzero class c , the locus $\{c \cdot \sigma = 0\}$ is a proper hypersurface of Q , so $\operatorname{NS}(\mathbb{T}_b) = 0$ off countably many hypersurfaces. On the other hand, every $\sigma \in Q$ is orthogonal to some real class c with $c^2 > 0$, since the orthogonal complement of the positive plane $\langle \operatorname{Re} \sigma, \operatorname{Im} \sigma \rangle$ has signature $(1, 3)$; approximating c by rational classes shows that the loci $\{c \cdot \sigma = 0\}$ with c rational and $c^2 > 0$ are dense in Q , and over them $\mathbb{T}_b$ is projective [4, Theorem IV.6.2].

Blowing up the zero section, we obtain a holomorphic family $\{X_b\}_{b \in B}$, where X_b is the blow-up of $\mathbb{T}_b$ at a point. Every X_b is Oka by [8, Corollary 6.4.13]. If $\mathbb{T}_b$ is projective, then so is X_b , and hence X_b is elliptic by [10] (or by Theorem 1.1). If $\operatorname{NS}(\mathbb{T}_b) = 0$, then $\mathbb{T}_b$ has algebraic dimension zero (Remark 2.11), and hence X_b is not elliptic by Example 2.9. Thus both the elliptic and the non-elliptic fibres occur over dense subsets of B , so ellipticity is neither open nor closed in this family.

3. GLOBALIZATION OF LOCAL SPRAYS

Let $f: U \rightarrow Y$ be a holomorphic map between complex manifolds. A *product spray* with *core* f is a holomorphic map $H: U \times P \rightarrow Y$ such that $H(\cdot, 0) = f$, where P is a ball centred at 0 in some $\mathbb{C}^M$; thus a product spray is a local spray over f on the trivial bundle $U \times \mathbb{C}^M$ that is defined on $U \times P$. It is said to be *dominating* over $S \subset U$ if $\partial_\zeta H(x, \zeta)$ maps onto $T_{Y, H(x, \zeta)}$ for all $x \in S$ and $\zeta \in P$. Such maps are called sprays of maps in [8, Chap. 5].

Theorem 3.1. *Let X be a weakly pseudoconvex manifold with a Kähler form ω_X , and let $E \rightarrow X$ be a vector bundle of rank $N \geq 1$ with a Hermitian metric. Assume that, for some smooth Hermitian metric on K_X^{-1} and some constant $c \geq 0$,*

$$(15) \quad \begin{aligned} \langle \sqrt{-1}\Theta_E(v, \bar{v})\xi, \xi \rangle &\leq -\omega_X(v, \bar{v})|\xi|^2 \quad (v \in T_{X, x}, \xi \in E_x, x \in X), \\ \sqrt{-1}\Theta(K_X^{-1} \otimes \det E) &\geq -c\omega_X. \end{aligned}$$

We identify E with its total space and let A be its zero section. Let Y be an Oka manifold, and let $f: E \rightarrow Y$ be a continuous map that is holomorphic on a neighbourhood U of A and admits a product spray with core $f|_U$ that is dominating over $U \setminus A$. Then, for every $r \geq 1$, there exists a holomorphic map $\tilde{f}: E \rightarrow Y$ that agrees with f to order r along A .

Theorem 1.4 is deduced from Theorem 3.1 in Section 4.

Strategy of the proof of Theorem 3.1. We follow the proof of the Oka principle for Stein sources [8, Chap. 5]. We exhaust the total space E by the disc bundles $\{\theta_w < 1\}$ defined below, and we pass from one disc bundle to a larger one in finitely many steps, each of which attaches a convex bump V to a set $U \supset A$ (Lemma 3.13). At each step, the current map F is holomorphic on U , and on the convex set V the Oka property of Y provides a holomorphic map G that approximates F on $U \cap V$. To glue F and G , we use dominating product sprays H with core F and T with core G . Near $U \cap V$, the old spray can be written as $H = T \circ \gamma$, where $\gamma(x, \cdot)$ is close to the identity map of the parameter space (Lemma 3.15). If $\gamma = \beta \circ \alpha^{-1}$ with α defined over U and β over V , then $H \circ \alpha$ and $T \circ \beta$ agree on $U \cap V$ and together form a product spray over $U \cup V$ (Lemma 3.16). This compositional splitting is a nonlinear version of the additive Cousin problem $c = b - a$, from which it is obtained by iteration (Lemma 3.7); the additive problem is solved via the $\bar{\partial}$ -equation (Lemma 3.6). Here U contains A and need not be compact or Stein, so we solve the $\bar{\partial}$ -equation with L^2 estimates for a weight q^{-m} that is singular along A ; this weight also forces α to agree with the identity to order r along A , which yields the interpolation along A . Since the disc bundles are not compact, we also need bounds near the ends of X (Lemmas 3.5 and 3.10). Finally, the functions θ_w have no critical points off A (Lemma 3.2(d)); hence, unlike in the Stein case, no handles need to be attached.

3.1. A splitting lemma on weakly pseudoconvex manifolds. In this subsection and in Subsection 3.2, let X , ω_X , E , c and the metric on K_X^{-1} be as in Theorem 3.1, and let $\rho \geq 0$ be a smooth psh exhaustion function of X . We identify E with its total space, denote by $\pi: E \rightarrow X$ the projection and by A the zero section, and set $n_E = \dim E$. Put

$$u(\xi) = |\xi|^2, \quad q = u e^u, \quad \Psi = \rho \circ \pi + u, \quad \theta_w = q e^{w \circ \pi} \quad (w \in C^\infty(X)),$$

and let $\mathcal{E}$ be the Euler vector field of E . We say that a holomorphic function *vanishes to order r along A* if all its derivatives of order $< r$ vanish on A .

The total space and L^2 estimates.

Lemma 3.2. (a) Ψ is a psh exhaustion function of E , and $\omega_E = \pi^* \omega_X + \sqrt{-1} \partial \bar{\partial} u$ is a Kähler form.

(b) $\sqrt{-1} \partial \bar{\partial} \log q \geq \omega_E$ on $E \setminus A$. Consequently, for a psh function w on X , $\log \theta_w$ is psh on E , θ_w is strictly psh on $E \setminus A$, and $-\log(b - \theta_w) + \Psi$ is a psh exhaustion function of $\{\theta_w < b\}$.

(c) With the induced metric h_0 on $K_E^{-1} \cong \pi^*(K_X^{-1} \otimes \det E)$ and $m \geq c + 1$,

$$(16) \quad \sqrt{-1} \Theta(K_E^{-1}, h_0) + m \sqrt{-1} \partial \bar{\partial} \log q \geq \omega_E \quad \text{on } E \setminus A.$$

(d) $d\theta_w(\mathcal{E} + \bar{\mathcal{E}}) = 2(1 + u)\theta_w$; in particular, θ_w has no critical points on $E \setminus A$.

(e) In local frames, $q^{1/2}$ is locally Lipschitz and u is comparable to the squared Euclidean norm $|\xi|^2$.

Proof. At $\xi_0 \in E_{y_0}$, in a frame normal at y_0 and for a tangent vector (v, τ) , (15) gives

$$\begin{aligned} \sqrt{-1}\partial\bar{\partial}u(v, \tau) &\geq |\tau|^2 + \omega_X(v, \bar{v})|\xi_0|^2, \\ \sqrt{-1}\partial\bar{\partial}\log u(v, \tau) &\geq \frac{|\tau|^2|\xi_0|^2 - |\langle \tau, \xi_0 \rangle|^2}{|\xi_0|^4} + \omega_X(v, \bar{v}) \geq \omega_X(v, \bar{v}) \end{aligned}$$

for $\xi_0 \neq 0$. This gives (a) and $\sqrt{-1}\partial\bar{\partial}\log q = \sqrt{-1}\partial\bar{\partial}\log u + \sqrt{-1}\partial\bar{\partial}u \geq \omega_E$; the rest of (b) follows from $\sqrt{-1}\partial\bar{\partial}e^\phi \geq e^\phi\sqrt{-1}\partial\bar{\partial}\phi$ and the compactness of $\{\Psi \leq M + \log b, \theta_w \leq b - e^{-M}\}$. For (c), we use the exact sequence $0 \rightarrow \pi^*E \rightarrow T_E \rightarrow \pi^*T_X \rightarrow 0$, (15) and $\pi^*\omega_X \leq \omega_E$. For (d), the identities $\partial u(\mathcal{E}) = u$ and $\partial(w \circ \pi)(\mathcal{E}) = 0$ give $\partial \log \theta_w(\mathcal{E}) = 1 + u$. Finally, (e) is clear from $u = |h(y)^{1/2}\xi|^2$. $\square$

For $m \geq c + 1$, let $d\mu$ be the volume form of h_0 and put $d\mu_m = q^{-m}d\mu$. For a $\mathbb{C}^k$ -valued $(0, j)$ -form g on an open set $S \subset E$, identified with a K_E^{-1} -valued (n_E, j) -form, put $\|g\|_S^2 = \int_S |g|^2 d\mu_m$, where $|g|$ is measured with respect to ω_E .

Lemma 3.3. *Let $D \subset E$ be a weakly pseudoconvex open set, let $j \geq 1$, and let g be a $\bar{\partial}$ -closed $(0, j)$ -form on D with L^2_{loc} coefficients and $\|g\|_D < \infty$. Then the solution v of $\bar{\partial}v = g$ with minimal norm exists, depends linearly on g , and satisfies $\|v\|_D \leq \|g\|_D$.*

Proof. $D \setminus A$ is complete Kähler. Let $\Phi \geq 0$ be a smooth psh exhaustion function of D . Then $\hat{\omega}_D = \omega_E + \sqrt{-1}\partial\bar{\partial}(\Phi^2)$ is a complete Kähler form on D : indeed, $\sqrt{-1}\partial\bar{\partial}(\Phi^2) \geq 2\sqrt{-1}\partial\bar{\partial}\Phi \wedge \bar{\partial}\Phi$, so the exhaustion function Φ has bounded gradient. Let $\eta: \mathbb{R} \rightarrow \mathbb{R}$ be smooth, convex and increasing, with $\eta(t) = -\log(-t)$ for $t \leq -1$. Since $\log q$ is psh (Lemma 3.2(b)), the form $\hat{\omega} = \hat{\omega}_D + \sqrt{-1}\partial\bar{\partial}\eta(\log q)$ is Kähler on $D \setminus A$. On the set where $\log q \leq -1$, we have $\sqrt{-1}\partial\bar{\partial}\eta(\log q) \geq \sqrt{-1}\partial\bar{\partial}\tau \wedge \bar{\partial}\tau$ with $\tau = \log(-\log q)$, and $\tau \rightarrow \infty$ at A ; hence $\hat{\omega}$ is complete on $D \setminus A$.

Solution. On $D \setminus A$, the metric h_0q^{-m} of K_E^{-1} has curvature $\geq \omega_E$ by (16), so the curvature operator on K_E^{-1} -valued (n_E, j) -forms is $\geq j \geq 1$ with respect to ω_E . Demailly's L^2 existence theorem [6, Théorème 4.1], which applies on the complete Kähler manifold $D \setminus A$ with estimates for the (not necessarily complete) Kähler form ω_E , gives v on $D \setminus A$ such that $\bar{\partial}v = g$ and $\|v\|_{D \setminus A} \leq \|g\|_D$. Since q is bounded on compact sets, v has L^2_{loc} coefficients on D , and the equation $\bar{\partial}v = g$ extends across the analytic set A [6, Lemme 6.9]. The minimal solution is the orthogonal projection of any solution onto the orthogonal complement of $\ker \bar{\partial}$. $\square$

Lemma 3.4. *If $m \geq N + r - 1$ and a is holomorphic on an open set $S \subset E$ and satisfies $\int_{S'} |a|^2 d\mu_m < \infty$ for all $S' \Subset S$, then a vanishes to order r along $A \cap S$.*

Proof. Locally, $d\mu_m$ is comparable to $|t|^{-2m}dV$, where t is the fibre coordinate. In the expansion $a = \sum_\beta a_\beta(x)t^\beta$, the monomials are orthogonal, and $\int |t^\beta|^2 |t|^{-2m}dV(t) = \infty$ if $|\beta| \leq r - 1$. $\square$

Fix a continuous function $C_E: E \rightarrow [1, \infty)$ such that the ω_E -ball $B(x, 4/C_E(x))$ lies in a chart in which ω_E , the Euclidean metric and $d\mu/dV$ are comparable with constant $C_E(x)$, and such that $q^{1/2}$ is $C_E(x)$ -Lipschitz on $B(x, 1/C_E(x))$.

Lemma 3.5. *There exists a continuous function $\kappa: E \rightarrow [1, \infty)$ with the following property. Let $R: E \rightarrow (0, 1]$ be a C^1 function, put $W_R = \{q < R^2\}$, and let $0 < \delta \leq \frac{1}{2}$, $\xi \in W_{(1-\delta)R}$ and $\Lambda \geq 2C_E(\xi)$. If $|d \log R| \leq \Lambda$ on $B(\xi, 2/\Lambda)$ and f is holomorphic on $W_R \cap B(\xi, 2/\Lambda)$ with norm $\leq b$ there, then*

$$|f(\xi)| \leq \kappa(\xi) (C_E(\xi) + \Lambda)^{n_E} \delta^{-n_E} R(\xi)^{m-n_E} b.$$

Proof. Put $t = \frac{\delta}{4} \min\{R(\xi)/C_E(\xi), 1/\Lambda\}$. For $y \in B(\xi, t)$, we have $q(y)^{1/2} < (1 - \frac{3}{4}\delta)R(\xi)$ and $R(y) \geq (1 - \frac{\delta}{4})R(\xi)$, so $B(\xi, t) \subset W_R \cap B(\xi, 2/\Lambda)$ and $q \leq R(\xi)^2$ there. The mean value inequality on a Euclidean ball of radius $t/C_E(\xi)$ gives $|f(\xi)| \leq C(\xi)t^{-n_E}R(\xi)^mb$, and $t^{-1} \leq 4\delta^{-1}(C_E(\xi) + \Lambda)R(\xi)^{-1}$. $\square$

Splitting.

Lemma 3.6 (Additive splitting). *Let $m \geq \max\{c+1, N+r-1\}$, and let $D = D_A \cup D_B \subset E$ be weakly pseudoconvex, where D_A, D_B are open, $\overline{D_B}$ is compact and disjoint from A , and $D_C = D_A \cap D_B$. Let $\chi \in C^\infty(E, [0, 1])$ satisfy $\chi = 0$ near $D_A \setminus D_B$ and $\chi = 1$ near $D_B \setminus D_A$, and assume that $\text{supp } d\chi$ is compact, disjoint from A , and satisfies $\text{supp } d\chi \cap D \subset D_C$. For a bounded holomorphic map $c: D_C \rightarrow \mathbb{C}^k$, let v_c be the minimal solution of $\bar{\partial}v = c\bar{\partial}\chi$ on D , and put $a_c = v_c - \chi c$ on D_A and $b_c = v_c + (1 - \chi)c$ on D_B . Then a_c and b_c are holomorphic, $b_c - a_c = c$, a_c vanishes to order r along A , and*

$$\|a_c\|_{D_A} + \|b_c\|_{D_B} \leq C_0 \sup |c|, \quad C_0 = 2\mu_m(D_B)^{1/2} + 2\|\bar{\partial}\chi\|_D,$$

where $\mu_m(D_B) = \int_{D_B} d\mu_m < \infty$. Moreover, holomorphic dependence on parameters is preserved.

Proof. The form $c\bar{\partial}\chi$ on D , extended by 0, is smooth and $\bar{\partial}$ -closed, since $\text{supp } d\chi \cap D \subset D_C$. Its norm is at most $\sup |c| \|\bar{\partial}\chi\|_D$, which is finite because $\text{supp } d\chi$ is compact and disjoint from A , so that q is bounded below on it. Similarly, χc and $(1 - \chi)c$ extend from D_C to D_A and to D_B , respectively, and are supported in D_B , so their norms are at most $\sup |c| \mu_m(D_B)^{1/2}$; here $\mu_m(D_B) < \infty$ because $\overline{D_B}$ is compact and disjoint from A . Lemma 3.3 gives $\|v_c\|_D \leq \|c\bar{\partial}\chi\|_D$, and the estimate follows. Near A , we have $a_c = v_c$, so a_c vanishes to order r along A by Lemma 3.4. The minimal solution depends linearly and boundedly on the data, which gives holomorphic dependence on parameters. $\square$

The next lemma is the analogue of the splitting lemma [8, Proposition 5.8.1] (a generalization of Cartan's lemma on invertible matrices) and of [20, Theorem 5.2]. It is purely formal: the geometry enters only through the hypothesis on additive splittings, which is verified for the pairs (U_λ, V_λ) in Lemma 3.14. Since the weighted L^2 estimates give supremum bounds only on slightly smaller sets, we replace the implicit function theorem on a fixed Banach space used in [8, 20] by a Newton iteration on shrinking domains.

Lemma 3.7 (Splitting lemma). *Let U_j, V_j ($j \geq 0$) be decreasing sequences of open sets with $V_0 \cap A = \emptyset$, put $C_j = U_j \cap V_j$, and let $U_\infty \subset \bigcap U_j$ and $V_\infty \subset \bigcap V_j$ be open. Let $\rho_j = \rho_\infty + (\rho_0 - \rho_\infty)2^{-j}$, and let $B_j \subset \mathbb{C}^k$ be the ball of radius ρ_j centred at 0. Assume that every bounded holomorphic map c on $C_j \times B_j$ splits as $c = b - a$ on $C_{j+1} \times B_j$, where a is defined on $U_{j+1} \times B_j$ and b on $V_{j+1} \times B_j$, $\sup |a| + \sup |b| \leq C2^{pj} \sup |c|$ ($C \geq 1$,*

$p \geq 0$), and $a(\cdot, \zeta)$ vanishes to order r along A . If γ is holomorphic on $C_0 \times B_0$ and $\epsilon_0 = \sup |\gamma - \zeta| \leq \epsilon_* := (\rho_0 - \rho_\infty)/(12C2^{p+1})$, then there exist holomorphic maps $\alpha: U_\infty \times B_\infty \rightarrow B_0$ and $\beta: V_\infty \times B_\infty \rightarrow \mathbb{C}^k$ such that $\gamma(x, \alpha(x, \zeta)) = \beta(x, \zeta)$ on $(U_\infty \cap V_\infty) \times B_\infty$, $\sup |\alpha - \zeta| + \sup |\beta - \zeta| \leq C'\epsilon_0$ with $C' = C'(C, p, \rho_0, \rho_\infty)$, and $\alpha(\cdot, \zeta) - \zeta$ vanishes to order r along A .

Proof. Put $s_j = \rho_j - \rho_{j+1}$ and $\gamma_0 = \gamma$. Given γ_j on $C_j \times B_j$ with $\epsilon_j = \sup |\gamma_j - \zeta|$, we split $\gamma_j - \zeta = b_j - a_j$, put $\eta_j = C2^{pj}\epsilon_j$, $\alpha_j = \zeta + a_j$ and $\beta_j = \zeta + b_j$, and assume that $\eta_j \leq s_j/8$. By the Cauchy estimates, $|\partial_\zeta a_j|, |\partial_\zeta b_j| \leq \frac{1}{4}$ on the ball of radius $\rho_{j+1} + s_j/2$; hence β_j^{-1} exists on the ball of radius $\rho_{j+1} + s_j/4$ by the contraction principle, and $\gamma_{j+1} = \beta_j^{-1} \circ \gamma_j \circ \alpha_j$ is defined on $C_{j+1} \times B_{j+1}$. Writing $\gamma_j \circ \alpha_j = \zeta + b_j + Q$ with $|Q| \leq 2\epsilon_j\eta_j/s_j$, we obtain $\epsilon_{j+1} \leq \frac{8}{3}\epsilon_j\eta_j/s_j \leq K2^{(p+1)j}\epsilon_j^2$, where $K = 6C/(\rho_0 - \rho_\infty)$. Hence $d_j = K2^{(p+1)(j+1)}\epsilon_j$ satisfies $d_{j+1} \leq d_j^2$ and $d_0 \leq \frac{1}{2}$, so $d_j \leq 2^{-2^j}$ and $\eta_j \leq s_j/12$, which keeps the induction going. Since $\sum \eta_i/s_i < \infty$, the compositions $\alpha_0 \circ \dots \circ \alpha_j$ and $\beta_0 \circ \dots \circ \beta_j$ have uniformly bounded Lipschitz constants, and they converge to maps α, β with the stated properties, since $\gamma \circ \alpha_0 \circ \dots \circ \alpha_j = \beta_0 \circ \dots \circ \beta_j \circ \gamma_{j+1}$ and $\gamma_j \rightarrow \zeta$. By the chain rule, vanishing to order r along A is preserved under composition with the α_i and under locally uniform limits. $\square$

3.2. Jet extension theorem from negative bundles to Oka manifolds. We construct the product sprays needed in Theorem 3.1.

Lemma 3.8. *Let $V \rightarrow U$ be a vector bundle with a section σ , put $A = \sigma^{-1}(0)$, and let s_0 be a dominating local spray over $f: U \rightarrow Y$ on a bundle $P_0 \rightarrow U$. Let $\tau_1, \dots, \tau_M$ be sections of $\text{Hom}(V, P_0)$ spanning it at every point, and let P be the unit ball of $\mathbb{C}^M$. Then, for some neighbourhood U' of A ,*

$$H(x, \zeta) = s_0\left(\sum_k \zeta_k \tau_k(x)(\sigma(x))\right), \quad x \in U', \quad \zeta \in P,$$

is a product spray with core $f|_{U'}$ that is dominating over $U' \setminus A$.

Proof. The points η at which the fibre derivative of s_0 is surjective form a neighbourhood Ω' of the zero section. Choose metrics on V and P_0 and a continuous function $\delta > 0$ such that $\{|\eta| < \delta(p(\eta))\} \subset \Omega'$, and let U' be the set where $(\sum_k |\tau_k|^2)^{1/2}|\sigma| < \delta$. For $x \in U'$ and $\zeta \in P$, the point $\psi_x(\zeta) = \sum_k \zeta_k \tau_k(x)(\sigma(x))$ lies in Ω' by the Cauchy–Schwarz inequality, and $\partial_\zeta H = d(s_0|_{P_{0,x}})_{\psi_x(\zeta)} \circ \psi_x$. If $\sigma(x) \neq 0$, then $\tau \mapsto \tau(\sigma(x))$ maps $\text{Hom}(V_x, P_{0,x})$ onto $P_{0,x}$, so ψ_x is surjective. $\square$

Lemma 3.9. *Let $L \rightarrow X$ be a Hermitian line bundle, let $E = L \otimes \mathbb{C}^N$ with projection π and zero section A , and let g be a holomorphic map from a neighbourhood W of A to Y such that $d(g|_{E_y})_{0_y}$ is surjective for all y . Let P be the unit Hilbert–Schmidt ball of $\mathbb{C}^{N \times N}$. Then, for some neighbourhood $U \subset W$ of A , the map $H(\xi, \Lambda) = g(\xi + \Lambda\xi)$ on $U \times P$ is a product spray with core $g|_U$ that is dominating over $U \setminus A$.*

Proof. We apply Lemma 3.8 on W with $V = P_0 = \pi^*E$, the tautological section $\sigma(\xi) = \xi$, the local spray $s_0(\xi, \eta) = g(\xi + \eta)$ over g , whose fibre derivative $d(g|_{E_{\pi(\xi)}})_\xi$ is surjective after shrinking W , and the matrix units $\tau_{ij} = e_i e_j^T$, which span $\pi^* \text{End } E \cong W \times \mathbb{C}^{N \times N}$ and satisfy $\sum \Lambda_{ij} \tau_{ij}(\xi) = \Lambda\xi$. $\square$

Disc bundles and convex bumps. Fix $r \geq 2$ such that $m = n_E + r - 1 \geq c + 1$ (so that $m \geq N + r - 1$), fix an open neighbourhood U of A , and put $L_j = \{\Psi \leq j\}$.

Lemma 3.10. *There exists a smooth convex increasing function $\chi: [0, \infty) \rightarrow [5, \infty)$ with $\chi' \rightarrow \infty$ such that*

- (a) $\theta_{\chi \circ \rho} \geq 2$ on $E \setminus U$;
- (b) if $\alpha \in [\frac{1}{2}, 2]$, $R = \alpha e^{-\chi(\rho \circ \pi)/2}$, $\rho(\pi(x)) \geq 1$ and $q(x) \leq 1$, then $R < 1$, and there exists $\Lambda \geq 2C_E(x)$ such that $|d \log R| \leq \Lambda$ and $\rho \circ \pi \geq \rho(\pi(x)) - 1$ on $B(x, 2/\Lambda)$ and $\kappa(x)(C_E(x) + \Lambda)^{n_E} R(x) \leq 1$.

Proof. Let $Z_t = \{\rho \circ \pi \leq t + 2, q \leq 4\}$, which is compact. Choose a continuous increasing function $C(t) \geq 1$ and a continuous decreasing function $\mu(t) \in (0, 1]$ such that $|d(\rho \circ \pi)|, C_E, \kappa \leq C(t)$ on Z_t and $q \geq \mu(t)$ on $\{\rho \circ \pi \leq t\} \setminus U$. Put $b = \max\{5, \log 2 - \log \mu\}$, $\Gamma = 16C^3$, $\nu = e^{-b/(2n_E)}$, $s(t) = \min\{(2n_E \Gamma(t+1))^{-1}, 2^{-t-1} \nu(t+1)\}$, choose a smooth strictly convex function p with $p \geq -\log s$, $p' \geq 1$ and $p(t) \geq t^2$, and put $\tau = \int_t^\infty e^{-p}$ and $\chi = -2n_E \log \tau$. By strict convexity, $\tau < e^{-p/p'}$; hence $\chi' = 2n_E e^{-p} e^{\chi/(2n_E)} \in [2n_E p', e^{\chi/(2n_E)}/\Gamma]$, so $\chi' \rightarrow \infty$. Moreover, $0 < \chi'' \leq (\chi')^2/(2n_E)$, and $\chi \geq b$ since $\tau(t) \leq \sum_{j \geq [t]} s(j) \leq \nu(t)$.

(a) Off U , we have $q \geq \mu \geq 2e^{-b} \geq 2e^{-\chi}$. (b) Let $t_x = \rho(\pi(x))$, $C = C(t_x)$ and $\Lambda = 4C^2 \chi'(t_x) \geq 2C_E(x)$. Along curves from x of length $< 2/\Lambda \leq 1/(2C \chi'(t_x))$ that stay in Z_{t_x} , the function $\rho \circ \pi$ changes by less than $1/(2\chi'(t_x)) \leq \frac{1}{2}$ and $q^{1/2} \leq \frac{3}{2}$, so such curves never leave Z_{t_x} . Since $(1/\chi')' \geq -\frac{1}{2}$, we have $\chi' \leq 2\chi'(t_x)$ on $[t_x - 1, t_x + 1/\chi'(t_x)]$, so $|d \log R| \leq C \chi'(t_x) \leq \Lambda$ on $B(x, 2/\Lambda)$. Finally, $\kappa(C_E + \Lambda)^{n_E} R \leq C(8C^2 \chi')^{n_E} 2e^{-\chi/2} \leq 2C(8C^2/\Gamma)^{n_E} \leq 1$. $\square$

For $T \geq 1$, we consider functions r on E satisfying

$$(17) \quad r = \theta_{\chi \circ \rho} - 1 \quad \text{on } \pi^{-1}(\{\rho \geq T\}).$$

Definition 3.11. A *convex bump* consists of smooth psh functions $r_D \leq r_U$ on E satisfying (17), with $r_D \geq -2$, $r_U = -1$ on A , and $\{r_D \leq 1\} \cap \pi^{-1}(K)$ compact for every compact set $K \subset X$, together with a chart $z: \Omega \rightarrow \{|z| < 3\}$, where $\Omega \subset \pi^{-1}(\{\rho < T + 1\}) \setminus A$, in which r_U, r_D are strictly convex, such that $r_U = r_D$ outside $\{|z| < 1\}$. For $0 < \lambda \leq \frac{1}{4}$, put $U_\lambda = \{r_U < \lambda\}$, $D_\lambda = \{r_D < \lambda\}$, $V_\lambda = D_\lambda \cap \{|z|^2 < 4 + \lambda\}$ and $C_\lambda = U_\lambda \cap V_\lambda$, and define $\chi_\Omega = h(|z|^2)$ on Ω and $\chi_\Omega = 0$ on $E \setminus \Omega$, where $h \in C^\infty(\mathbb{R}, [0, 1])$ satisfies $h = 1$ on $(-\infty, 1]$ and $h = 0$ on $[2, \infty)$.

The pair (U_λ, V_λ) plays the role of a Cartan pair [8, Definition 5.7.1] or of a 1-convex Cartan pair [20, Definition 5.1]. Of the defining properties of a Cartan pair, we use only the separation property (Lemma 3.12(c)), which gives the cut-off function in the additive splitting, and the convexity of V_λ and C_λ (Lemma 3.12(a)), which is needed for approximation and for transition maps. The Stein property of D_λ , which in [8, Lemma 5.8.2] yields solutions of the $\bar{\partial}$ -equation with bounds, is replaced by the weak pseudoconvexity of D_λ (Lemma 3.12(b)) and the weighted L^2 estimates of Lemma 3.3; as in [20, Proposition 4.5], the $\bar{\partial}$ -data are supported away from A .

- Lemma 3.12.** (a) $D_\lambda = U_\lambda \cup V_\lambda$; V_λ and $C_\lambda = U_\lambda \cap \{|z|^2 < 4 + \lambda\}$ are convex in the chart; $\overline{V_\lambda} \subset \Omega$ is compact.
 (b) U_λ and D_λ are weakly pseudoconvex.
 (c) On D_λ : $\{\chi_\Omega < 1\} \subset U_\lambda$, $\{\chi_\Omega > 0\} \subset V_\lambda$, $\text{supp } d\chi_\Omega \subset C_\lambda$.
 (d) There exists $c_1 > 0$ such that, for $\lambda' < \lambda$, the sets $V_{\lambda'}$ and $U_{\lambda'} \cap \pi^{-1}(\{\rho \leq T + 1\})$ have distance $\geq c_1(\lambda - \lambda')$ from $E \setminus V_\lambda$ and $E \setminus U_\lambda$.

Proof. Assertions (a) and (c) follow since $r_D < r_U$ only in $\{|z| < 1\}$ and $\text{supp } d\chi_\Omega \subset \{1 \leq |z|^2 \leq 2\}$. For (b), use $-\log(\lambda - r) + \Psi$ as in Lemma 3.2(b). For (d), bound $|dr_U|, |dr_D|, |d|z|^2|$ on a neighbourhood of the compact set $(\{r_D \leq 1\} \cap \pi^{-1}(\{\rho \leq T + 1\})) \cup \{|z|^2 \leq \frac{17}{4}\}$. $\square$

Lemma 3.13 (Decomposition into convex bumps). *Let $T \geq 1$, let $w \geq w'$ be smooth psh functions on X with $w = w' = \chi \circ \rho$ on $\{\rho \geq T\}$, put $r_s = \theta_{(1-s)w+sw'} - 1$, and let $0 < \epsilon \leq \frac{1}{4}$. Then there exist convex bumps $(r_U^{(i)}, r_D^{(i)})$, $i = 1, \dots, I$, with charts in $\{\theta_w > 1 - 2\epsilon\}$, such that $r_1 \leq r_D^{(i)} \leq r_U^{(i)} \leq r_0$ and, for $|\lambda| \leq \epsilon/2$,*

$$\{r_U^{(1)} < \lambda\} = \{r_0 < \lambda\}, \quad \{r_D^{(i)} < \lambda\} = \{r_U^{(i+1)} < \lambda\}, \quad \{r_D^{(I)} < \lambda\} = \{r_1 < \lambda\}.$$

Proof. This is the usual construction for noncritical strongly psh exhaustion functions [8, Lemma 5.10.3], carried out over $\{\rho \leq T\}$. By Lemma 3.2, the functions r_s are psh on E , strictly psh and without critical points on $E \setminus A$, equal to -1 on A , and independent of s over $\{\rho \geq T\}$. Since $r_s + 1 = q e^{((1-s)w+sw') \circ \pi}$ and $w' \leq w$, they are nonincreasing in s , and $r_{s'} \rightarrow r_s$ in C^2 on compact sets as $s' \rightarrow s$. (They are not affine in s ; only $\log(r_s + 1)$ is.)

Charts. The set Z of pairs $(s, x) \in [0, 1] \times E$ with $\rho(\pi(x)) \leq T$ and $|r_s(x)| \leq \epsilon$ is compact and does not meet $[0, 1] \times A$. By Narasimhan's lemma, for every $(s_*, x_*) \in Z$ there exist a chart $z: \Omega \rightarrow \{|z| < 3\}$ centred at x_* , with $\Omega \subset \{\theta_w > 1 - 2\epsilon\} \cap \pi^{-1}(\{\rho < T + 1\}) \setminus A$, and a number $\delta > 0$ such that r_s is strictly convex in z near $\overline{\Omega}$ for $|s - s_*| < 2\delta$. Finitely many of the sets $(s_* - \delta, s_* + \delta) \times \{|z| < 1\}$ cover Z ; let δ_0 be the smallest of the corresponding δ . Cover $[0, 1]$ by closed intervals $I_1, \dots, I_p$ of length $< \delta_0$. For each a , let $\Omega_{a,1}, \dots, \Omega_{a,K_a}$ be the charts of this cover whose interval meets I_a . Then r_s is strictly convex near each $\Omega_{a,k}$ for $s \in I_a$, and the balls $\{|z| < 1\}$ of these charts cover $M_a = \{x : (s, x) \in Z \text{ for some } s \in I_a\}$. Choose smooth functions $\chi_{a,k} \geq 0$ supported in these balls such that $\sum_k \chi_{a,k} \leq 1$, with equality near M_a .

Bumps. Since there are only finitely many functions $\chi_{a,k}$ and $r_{s'} \rightarrow r_s$ in C^2 on compact sets as $s' \rightarrow s$, there exists a subdivision $0 = s_0 < s_1 < \dots < s_L = 1$ such that each $[s_l, s_{l+1}]$ lies in some $I_{a(l)}$ and the functions

$$r_{l,k} = r_{s_l} + \sum_{k' \leq k} \chi_{a(l),k'}(r_{s_{l+1}} - r_{s_l}), \quad 0 \leq k \leq K_{a(l)},$$

are so close to r_{s_l} in C^2 on $\bigcup_k \overline{\Omega_{a(l),k}}$ that they are strictly convex on each $\Omega_{a(l),k}$. Outside these charts, $r_{l,k} = r_{s_l}$; hence $r_{l,k}$ is psh on E . At each point, $r_{l,k}$ is a convex combination of r_{s_l} and $r_{s_{l+1}}$, so $r_1 \leq r_{l,k} \leq r_0$, $r_{l,k} = -1$ on A , and (17) holds. The difference $r_{l,k-1} - r_{l,k} \geq$

0 is supported in the ball $\{|z| < 1\}$ of $\Omega_{a(l),k}$. Therefore the pairs $(r_U, r_D) = (r_{l,k-1}, r_{l,k})$, $1 \leq k \leq K_{a(l)}$, with the charts $\Omega_{a(l),k}$, ordered lexicographically in (l, k) , are convex bumps.

Matching. Consecutive bumps with the same l share a function, and $r_{0,0} = r_0$. It remains to compare $r_{l,K}$, where $K = K_{a(l)}$, with $r_{l+1,0} = r_{s_{l+1}}$. They agree near $M_{a(l)}$ and over $\{\rho > T\}$. At any other point x , we have $|r_s(x)| > \epsilon$ for all $s \in [s_l, s_{l+1}]$, so $r_{s_l}(x)$ and $r_{s_{l+1}}(x)$ have the same sign and absolute values $> \epsilon$, and $r_{l,K}(x)$ lies between them. Hence $\{r_{l,K} < \lambda\} = \{r_{s_{l+1}} < \lambda\}$ for $|\lambda| \leq \epsilon/2$. $\square$

Lemma 3.14 (Splitting over a convex bump). *Let (r_U, r_D) be a convex bump, let $\frac{1}{4} \geq \lambda_+ > \lambda_- > 0$, and let $B_0, B_\infty \subset \mathbb{C}^k$ be balls centred at 0 with radii $\rho_0 > \rho_\infty$. Then there exist $\epsilon_*, C' > 0$ such that, for every holomorphic map γ on $C_{\lambda_+} \times B_0$ with $\sup |\gamma - \zeta| \leq \epsilon_*$, there exist holomorphic maps $\alpha: U_{\lambda_-} \times B_\infty \rightarrow B_0$ and $\beta: V_{\lambda_-} \times B_\infty \rightarrow \mathbb{C}^k$ such that $\gamma(x, \alpha(x, \zeta)) = \beta(x, \zeta)$ on $C_{\lambda_-} \times B_\infty$, $\sup |\alpha - \zeta| + \sup |\beta - \zeta| \leq C' \sup |\gamma - \zeta|$, and $\alpha(\cdot, \zeta) - \zeta$ vanishes to order r along A .*

Proof. Let $\lambda_j = \lambda_- + (\lambda_+ - \lambda_-)2^{-j}$. We verify the hypothesis of Lemma 3.7 for the sets $U_{\lambda_j}, V_{\lambda_j}$ with $p = n_E$. Applying Lemma 3.6 on D_{λ_j} with $D_A = U_{\lambda_j}$, $D_B = V_{\lambda_j}$ and the cut-off function χ_Ω (Lemma 3.12), we split c with L^2 norms $\leq C_0 \sup |c|$, where $C_0 = 2\mu_m(V_{\lambda_+})^{1/2} + 2\|\bar{\partial}\chi_\Omega\|_E$ does not depend on j since $V_{\lambda_j} \subset V_{\lambda_+}$ and $\overline{V_{\lambda_+}}$ is a compact subset of $\Omega \subset E \setminus A$. On $V_{\lambda_{j+1}}$ and on $U_{\lambda_{j+1}} \cap \pi^{-1}(\{\rho \leq T+1\})$, the mean value inequality on balls of radius $c_1(\lambda_j - \lambda_{j+1})$ (Lemma 3.12(d)) gives sup bounds of the form $O(2^{n_E j}) \sup |c|$. At a point $\xi \in U_{\lambda_{j+1}}$ with $\rho(\pi(\xi)) \geq T+1$, we have $U_{\lambda_j} = W_{R_j}$ near ξ , where $R_j = (1 + \lambda_j)^{1/2} e^{-\chi(\rho \circ \pi)/2}$, as well as $\xi \in W_{(1-\delta_j)R_j}$ with $\delta_j \geq (\lambda_j - \lambda_{j+1})/4$ and $q(\xi) \leq 1$. Lemma 3.10(b) gives Λ with $B(\xi, 2/\Lambda) \subset \pi^{-1}(\{\rho \geq T\})$, and Lemma 3.5, with $m - n_E \geq 1$, then gives $|a_c(\xi)| \leq C_0 \delta_j^{-n_E} \sup |c|$. $\square$

Gluing. Let Y be an Oka manifold with a complete Riemannian metric, and let $P_s \subset \mathbb{C}^M$ denote the ball of radius s . We use the basic Oka property with approximation for maps from Stein manifolds to Y [8, Corollary 5.4.5], the polynomial convexity of compact convex sets [12], the existence of dominating sprays over holomorphic maps from Stein manifolds [8, Lemma 5.10.4], and the homotopy extension property of $(E, \{r \leq \lambda\})$ for regular values λ . As in [8, Sects. 5.9–5.10] and [20, Sects. 6 and 8], we first glue two sprays over a pair (U_λ, V_λ) and then extend across a convex bump.

Lemma 3.15 (Transition maps; [8, Lemma 5.9.3], [20, Lemma 6.2]). *Let S be a Stein manifold, let $G: S \times P_1 \rightarrow Y$ be a holomorphic map such that $\partial_\zeta G$ is surjective everywhere, let $L \subset S$ be compact, and let L_1 be a compact neighbourhood of L . Then, for every $\epsilon_1 \in (0, \frac{1}{16}]$, there exist a neighbourhood W of L and $\delta_1 > 0$ with the following property: for every k and every holomorphic map T defined near $L_1 \times \overline{P_{3/4}} \times \mathbb{B}^k$ with $\text{dist}(T(x, \zeta, w), G(x, \zeta)) < \delta_1$ there, there exists a holomorphic map $\gamma_\zeta: W \times P_{1/4} \times \mathbb{B}^k \rightarrow P_{1/2}$ such that $|\gamma_\zeta - \zeta| < \epsilon_1$ and $T(x, \gamma_\zeta(x, \zeta, w), w) = G(x, \zeta)$.*

Proof. We follow [8, p. 240], keeping track of the parameter domain. Let Π be a holomorphic right inverse of $\partial_\zeta G$, which exists by Theorem B. By the inverse function theorem, $v \mapsto G(x, \zeta + \Pi v)$ has a local inverse $y \mapsto v(x, \zeta, y)$ near $G(x, \zeta)$, uniformly for (x, ζ) near

$L \times \overline{P_{1/2}}$. For sufficiently small δ_1 , independent of k and T , the map $\eta = \zeta + \Pi v(x, \zeta, T)$ satisfies $T = G(x, \eta)$ and $|\eta - \zeta| < \epsilon_1$; solving $\eta(x, \gamma_\zeta, w) = \zeta'$ by the contraction principle on $\overline{B}(\zeta', 2\epsilon_1)$ (using the Cauchy estimates) gives γ_ζ . $\square$

Lemma 3.16 (Gluing lemma; cf. [8, Proposition 5.9.2], [20, Proposition 6.1]). *Let (r_U, r_D) be a convex bump, let $\frac{1}{4} \geq \lambda_1 > \lambda_+ > \lambda_- > 0$ and $k \geq 0$, and let $H: U_{\lambda_1} \times P_1 \rightarrow Y$ be a product spray that is dominating over $U_{\lambda_1} \setminus A$. Then, for every $\epsilon' > 0$, there exists $\delta' > 0$ with the following property. Let T be a holomorphic map from a neighbourhood of $\overline{V_{\lambda_+}} \times \overline{P_{3/4}} \times \mathbb{B}^k$ to Y such that $\partial_{(\zeta, w)} T$ is surjective everywhere and $\text{dist}(T(x, \zeta, w), H(x, \zeta)) < \delta'$ on $\overline{C_{\lambda_+}} \times \overline{P_{3/4}} \times \mathbb{B}^k$. Then there exist holomorphic maps $\alpha = (\alpha_\zeta, \alpha_w): U_{\lambda_-} \times P' \rightarrow P_{1/2} \times \mathbb{B}^k$ and $\beta: V_{\lambda_-} \times P' \rightarrow P_{1/2} \times \mathbb{B}^k$, where $P' \subset \mathbb{C}^{M+k}$ is the ball of radius $\frac{1}{16}$, such that $\alpha(\cdot, v) - v$ vanishes to order r along A , $|\alpha(x, v) - v|, |\beta(x, v) - v| < \epsilon'$, and*

$$H'(x, v) = H(x, \alpha_\zeta(x, v)) \text{ on } U_{\lambda_-} \times P', \quad H'(x, v) = T(x, \beta(x, v)) \text{ on } V_{\lambda_-} \times P'$$

define a product spray $H': D_{\lambda_-} \times P' \rightarrow Y$ that is dominating over $D_{\lambda_-} \setminus A$.

Proof. Choose $\lambda' \in (\lambda_-, \lambda_+)$. Let ϵ_*, C' be the constants of Lemma 3.14 for the levels $\lambda' > \lambda_-$ and the balls of radii $\frac{1}{4}, \frac{1}{8}$ in $\mathbb{C}^{M+k}$, and put $\epsilon_1 = \min\{\epsilon_*, 1/(32C'), \epsilon'/(2C')\}$. Applying Lemma 3.15 to $G = H$ on the convex domain $S = C_{\lambda_1}$, which does not meet A , with $L = \overline{C_{\lambda'}}$ and $L_1 = \overline{C_{\lambda_+}}$, we obtain W and $\delta' = \delta_1$. Then $\gamma(x, v) = (\gamma_\zeta(x, v), w)$, where $v = (\zeta, w)$, is defined on $W \times P_{1/4} \times \mathbb{B}^k$, satisfies $|\gamma - v| < \epsilon_1$, and $T(x, \gamma(x, v)) = H(x, \zeta)$. Lemma 3.14 gives α on U_{λ_-} and β on V_{λ_-} such that $\gamma \circ \alpha = \beta$ on C_{λ_-} , $\sup |\alpha - v| + \sup |\beta - v| \leq C'\epsilon_1$, and $\alpha(\cdot, v) - v$ vanishes to order r along A . By the Cauchy estimates, $\partial_v \alpha$ and $\partial_v \beta$ are within $\frac{1}{2}$ of the identity on P' . On $C_{\lambda_-} \times P'$, we have $T \circ \beta = T \circ \gamma \circ \alpha = H(x, \alpha_\zeta)$, so H' is well defined; it is dominating by the chain rule, since H is dominating over $U_{\lambda_1} \setminus A$ and T is dominating. $\square$

Lemma 3.17 (Extension across a convex bump; cf. [8, Lemma 5.10.5], [20, Proposition 8.3]). *Let (r_U, r_D) be a convex bump, let $0 < \lambda_0 < \lambda_1 \leq \frac{1}{4}$, where λ_0 is a regular value of r_D , let $F: E \rightarrow Y$ be a continuous map, and let $H: U_{\lambda_1} \times P \rightarrow Y$ be a product spray with core $F|_{U_{\lambda_1}}$ that is dominating over $U_{\lambda_1} \setminus A$. Let $K \subset \{r_U < 0\}$ be compact and $\epsilon > 0$. Then there exist a continuous map $F': E \rightarrow Y$ that agrees with F to order r along A and satisfies $\text{dist}(F', F) < \epsilon$ on K , and a product spray $H': D_{\lambda_0} \times P' \rightarrow Y$, $P' \subset \mathbb{C}^{M+\dim Y}$, with core $F'|_{D_{\lambda_0}}$ that is dominating over $D_{\lambda_0} \setminus A$.*

Proof. We may assume that $P = P_1$. Put $k = \dim Y$, choose $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_0$, and put $V' = V_{\lambda_1}$. The sets $K_U = \overline{C_{\lambda_2}} \subset K_V = \overline{V_{\lambda_2}}$ are compact and convex in V' (Lemma 3.12(a)), and $\overline{V_{\lambda_3}} \subset K_V$.

Tolerances. Let 2δ be less than the convexity radius of Y near $F(K_U)$. Choose $\epsilon' > 0$ such that $\text{dist}(H(x, \zeta), F(x)) < \min\{\delta/2, \epsilon\}$ for $x \in K_U \cup K$ and $|\zeta| < \epsilon'$, let δ' be the constant given by Lemma 3.16 for $\lambda_1 > \lambda_2 > \lambda_3$, k and ϵ' , and put $\delta_1 = \min\{\delta', \delta/2\}$.

Approximation. Let $\hat{H}(x, \zeta) = H(x, \psi(x)\zeta)$, where $\psi \in C_c^\infty(U_{\lambda_1} \cap V')$ satisfies $0 \leq \psi \leq 1$ and equals 1 near K_U (so that $\psi(x)\zeta \in P_1$ for $\zeta \in P_1$), and let $\hat{H} = F$ elsewhere on $V' \times P_1$. The set $K_U \times \overline{P_{7/8}}$ is compact and convex, hence $\mathcal{O}(V' \times P_1)$ -convex. The basic

Oka property gives a homotopy $\hat{H}_t$ from $\hat{H}$ to a holomorphic map G on $V' \times P_1$ such that $\hat{H}_t$ is holomorphic near $K_U \times \overline{P_{7/8}}$ and satisfies $\text{dist}(\hat{H}_t, H) < \delta_1/2$ there; put $g_t = \hat{H}_t(\cdot, 0)$.

Thickening. We apply [8, Lemma 5.10.4] to the section $(x, \zeta) \mapsto ((x, \zeta), G(x, \zeta))$ of $(V' \times P_1) \times Y \rightarrow V' \times P_1$ and the compact convex set $K_V \times \overline{P_{3/4}}$. Since this set has contractible Stein neighbourhoods, the lemma gives a spray $\hat{G}(x, \zeta, w)$, with w near $0 \in \mathbb{C}^k$, such that $\hat{G}(x, \zeta, 0) = G(x, \zeta)$ and $\partial_w \hat{G}$ is surjective near $K_V \times \overline{P_{3/4}} \times \{0\}$. For sufficiently small $\sigma > 0$, the map $T(x, \zeta, w) = \hat{G}(x, \zeta, \sigma w)$ is defined near $K_V \times \overline{P_{3/4}} \times \overline{\mathbb{B}^k}$, $\partial_w T$ is surjective there, and $\text{dist}(T, H) < \delta_1$ on $K_U \times \overline{P_{3/4}} \times \overline{\mathbb{B}^k}$.

Gluing. Lemma 3.16 with $\lambda_+ = \lambda_2$ and $\lambda_- = \lambda_3$ gives α, β and a spray H'' over D_{λ_3} that is dominating off A . Its core is $f' = H(x, a(x))$ on U_{λ_3} , where $a = \alpha_\zeta(\cdot, 0)$, and $f' = T(x, b(x))$ on V_{λ_3} , where $b = \beta(\cdot, 0)$; moreover, $|a|, |b| < \varepsilon'$. Since a vanishes to order r along A , so does $f' - F = \int_0^1 \partial_\zeta H(x, \tau a) a d\tau$ in local coordinates, and $\text{dist}(f', F) < \varepsilon$ on $K \subset U_{\lambda_3}$.

Homotopy. On U_{λ_3} , let $\Phi_t^U = H(x, (1-t)a(x))$; on V_{λ_3} , let $\Phi_t^V = T(x, (1-2t)b(x))$ for $t \leq \frac{1}{2}$ and $\Phi_t^V = g_{2-2t}$ for $t \geq \frac{1}{2}$, which is continuous since $T(x, 0) = G(x, 0) = g_1(x)$. On $C_{\lambda_3} \subset K_U$, both homotopies lie within δ of $F(x)$ by the choice of ε' and δ_1 . Hence $\Phi_t = [\Phi_t^U, \Phi_t^V]_{\chi_\Omega}$, the point dividing the minimal geodesic in the ratio $\chi_\Omega : (1 - \chi_\Omega)$, is a homotopy on $\overline{D_{\lambda_0}}$ from f' to F (Lemma 3.12(c)). By the homotopy extension property, Φ_{1-t} extends to a homotopy on E starting at F , whose end map F' equals f' on $\overline{D_{\lambda_0}} \supset A \cup K$. Finally, put $H' = H''|_{D_{\lambda_0} \times P'}$. $\square$

The approximation theorem. The sublevel set $\{\theta_w < c\}$ is the disc bundle $\{|\xi| < R(\pi(\xi))\}$, whose radius R is determined by $q(R) = ce^{-w}$. The following approximation theorem enlarges the radius from about $e^{-w/2}$ to about $e^{-w'/2}$; it combines Lemmas 3.13 and 3.17 and corresponds to [8, Proposition 5.10.1].

Proposition 3.18. *Let $T \geq 1$, let $w \geq w'$ be smooth psh functions on X with $w = w' = \chi \circ \rho$ on $\{\rho \geq T\}$, and let $0 < \ell \leq \frac{1}{2}$. Let $F: E \rightarrow Y$ be a continuous map that is holomorphic on $W = \{\theta_w < 1 + \ell\}$ and admits a product spray with core $F|_W$ that is dominating over $W \setminus A$. Let $K \subset \{\theta_w < \frac{1}{2}\}$ be compact and $\varepsilon > 0$. Then there exist $\ell' \in (0, \frac{1}{2}]$ and a continuous map $F': E \rightarrow Y$ that is holomorphic on $W' = \{\theta_{w'} < 1 + \ell'\}$, admits a product spray with core $F'|_{W'}$ that is dominating over $W' \setminus A$, agrees with F to order r along A , and satisfies $\text{dist}(F', F) < \varepsilon$ on K .*

Proof. Lemma 3.13 with $\epsilon = \ell/2$ gives convex bumps $(r_U^{(i)}, r_D^{(i)})$, $i = 1, \dots, I$, with charts in $\{\theta_w > 1 - \ell\} \subset \{\theta_w > \frac{1}{2}\}$. By Sard's theorem, we can choose $\ell/4 \geq \lambda^{(0)} > \lambda^{(1)} > \dots > \lambda^{(I)} > 0$ such that $\lambda^{(i)}$ is a regular value of $r_D^{(i)}$ for $i \geq 1$. Put $F_0 = F$, and restrict the given spray to $\{r_0 < \lambda^{(0)}\} = \{\theta_w < 1 + \lambda^{(0)}\} \subset W$. For $i = 1, \dots, I$, we apply Lemma 3.17 to the i -th bump with $\lambda_1 = \lambda^{(i-1)}$, $\lambda_0 = \lambda^{(i)}$, the set K and ε/I ; this is possible since, by the matching of levels in Lemma 3.13, the previous spray is defined on $\{r_U^{(i)} < \lambda^{(i-1)}\}$, and $K \subset \{r_U^{(i)} < 0\}$ since $r_U^{(i)} \leq r_0 < -\frac{1}{2}$ on K . Finally, $\{r_D^{(I)} < \lambda^{(I)}\} = \{\theta_{w'} < 1 + \lambda^{(I)}\}$, so $F' = F_I$ and $\ell' = \lambda^{(I)}$ have the required properties. $\square$

To exhaust E , we use the following choice of weights.

Lemma 3.19. *There exist smooth convex nondecreasing functions $\chi = \varphi_0 \geq \varphi_1 \geq \dots$ and numbers $1 \leq t_1 < t_2 < \dots$ such that $\varphi_j = \chi$ on $[t_j, \infty)$ and $L_j \subset \{\theta_{\varphi_j \circ \rho} < \frac{1}{2}\}$.*

Proof. Let $\beta: \mathbb{R} \rightarrow [0, 1]$ be a smooth nondecreasing function with $\beta = 0$ on $(-\infty, -1]$ and $\beta = 1$ on $[0, \infty)$. For $t \geq 1$, put $k_t(\tau) = \chi''(\tau)(1 - \beta(\tau - t))$, $h_t(s) = \int_s^t \int_\sigma^t k_t$ (with $h_t = 0$ on $[t, \infty)$), and $\varphi^{(t)} = \chi - h_t$. Then $(\varphi^{(t)})'' = \chi''\beta(\cdot - t) \geq 0$, $(\varphi^{(t)})' = \chi' + \int_s^t k_t \geq 0$, $\varphi^{(t)} = \chi$ on $[t, \infty)$, $\varphi^{(t)}$ decreases in t , and for $s \leq t - 2$, $h_t(s) \geq \chi'(t - 1) - \chi'(s + 1) \rightarrow \infty$. We choose t_j inductively so large that $L_j \subset \{\theta_{\varphi^{(t_j)} \circ \rho} < \frac{1}{2}\}$, and put $\varphi_j = \varphi^{(t_j)}$. $\square$

Proof of Theorem 3.1. We may assume that $r \geq 2$ and $n_E + r - 1 \geq c + 1$. Let $w_j = \varphi_j \circ \rho$ (Lemma 3.19); then $w_j \geq w_{j+1}$ and $w_j = w_{j+1} = \chi \circ \rho$ on $\{\rho \geq t_{j+1}\}$. By Lemma 3.10(a), we have $\{\theta_{w_0} < \frac{3}{2}\} \subset U$; thus f is holomorphic on this set and admits the given spray there. Applying Proposition 3.18 successively to $w_j \geq w_{j+1}$, with $K = L_j$ ($K = \emptyset$ for $j = 0$) and $\varepsilon = 2^{-j}$, we obtain continuous maps $F_j: E \rightarrow Y$ with $F_0 = f$ that are holomorphic on $\{\theta_{w_j} < 1 + \ell_j\} \supset L_j$, agree with f to order r along A , and satisfy $\text{dist}(F_{j+1}, F_j) < 2^{-j}$ on L_j . Since every compact subset of E lies in some L_j , the maps F_j converge locally uniformly to a holomorphic map $\tilde{f}: E \rightarrow Y$, which agrees with f to order r along A by the Cauchy estimates. $\square$

4. PROOF OF GROMOV'S CONJECTURE

In this section, we first apply Corollary 2.7 to weakly pseudoconvex manifolds admitting a positive line bundle, and then prove Theorems 1.1 and 1.4.

Theorem 4.1. *Let X be a weakly pseudoconvex complex manifold admitting a positive line bundle. Then X admits a local dominating spray.*

Proof of Theorem 4.1. By Takayama's embedding theorem [21], there exists an embedding $\iota: X \hookrightarrow \mathbb{P}^N$, which is not necessarily proper. Set

$$H := \iota^* \mathcal{O}_{\mathbb{P}^N}(1) \quad \text{and} \quad B := K_X \otimes H^{\otimes(2n+1)},$$

where $n := \dim X$. We use the following claim.

Claim 4.2. *The line bundle $B = K_X \otimes H^{\otimes(2n+1)}$ has the following properties:*

- B is positive.
- B is generated by finitely many sections.
- Finitely many sections of B define an embedding of X into a projective space.
- $T_X \otimes B$ is generated by finitely many sections.

Proof of Claim 4.2. Since $\iota: X \hookrightarrow \mathbb{P}^N$ is an embedding, H is positive and generated by finitely many sections. The natural surjection $\iota^* \Omega_{\mathbb{P}^N}^1 \rightarrow \Omega_X^1$ induces a surjective bundle morphism

$$\iota^*(\Omega_{\mathbb{P}^N}^1(2)) \twoheadrightarrow \Omega_X^1 \otimes H^{\otimes 2}.$$

The Euler sequence on $\mathbb{P}^N$ shows that $\Omega_{\mathbb{P}^N}^1(2)$ is generated by the sections represented by $z_i dz_j - z_j dz_i$ for $0 \leq i < j \leq N$. Hence its quotient $\Omega_X^1 \otimes H^{\otimes 2}$ and all exterior powers of this quotient are generated by finitely many sections. This shows that

$$B \otimes H^\vee \cong \det(\Omega_X^1 \otimes H^{\otimes 2})$$

is generated by finitely many sections. In particular, $B \otimes H^\vee$ admits a metric with semi-positive curvature. Tensoring this metric with the positive metric on H , we conclude that B is positive. Products of generating sections of $B \otimes H^\vee$ and H generate B . Moreover, let ψ be the morphism defined by finitely many generating sections of $B \otimes H^\vee$. The products of these sections with the sections of H defining ι define the composition of (ι, ψ) with the Segre embedding. Since ι is an embedding, so is this composition. This proves the third property. By the canonical isomorphism $\bigwedge^{n-1} \Omega_X^1 \cong T_X \otimes K_X$, we have

$$T_X \otimes B \cong \bigwedge^{n-1} (\Omega_X^1 \otimes H^{\otimes 2}) \otimes H^{\otimes 3}.$$

Both factors on the right are generated by finitely many sections, which proves the last property. $\square$

Choose finitely many sections

$$\{\xi_i\}_{i \in I} \subset H^0(X, T_X \otimes B)$$

generating $T_X \otimes B$, and set $m := |I|$. Tensoring the corresponding evaluation morphism with $B^\vee$ gives a surjective bundle morphism

$$\varphi: E := (B^\vee)^{\oplus m} \longrightarrow T_X.$$

Since H is positive and X is weakly pseudoconvex, Nakano's vanishing theorem for weakly pseudoconvex manifolds [16, 17] gives

$$H^1(X, B) = H^1(X, K_X \otimes H^{\otimes(2n+1)}) = 0.$$

As $S^2 E^\vee \otimes E \cong B^{\oplus m \binom{m+1}{2}}$, it follows that $H^1(X, S^2 E^\vee \otimes E) = 0$, and hence $\alpha_\varphi(E) = 0$. Corollary 2.7 therefore yields a local dominating spray on E , completing the proof. $\square$

Proof of Theorem 1.4. Choose a metric h_B on B with positive curvature, and put $\omega_X = \sqrt{-1}\Theta_{h_B}$, which is a Kähler form. Equip $K_X^{-1} \otimes B^{\otimes k}$ with a metric of semipositive curvature, and K_X^{-1} with the induced metric; then $\sqrt{-1}\Theta(K_X^{-1}) \geq -k\omega_X$. Equip $E = (B^\vee)^{\oplus N}$ with the metric induced by h_B^{-1} . Then $\sqrt{-1}\Theta_E = -\omega_X \otimes \text{id}_E$, and

$$\sqrt{-1}\Theta(K_X^{-1} \otimes \det E) = \sqrt{-1}\Theta(K_X^{-1}) - N\omega_X \geq -(k+N)\omega_X.$$

Hence (15) holds with the constant $c = k + N$. We identify E with its total space, let $\pi: E \rightarrow X$ be the projection and A the zero section, and let $s: \Omega \rightarrow X$ be the given local dominating spray.

The map. Choose a continuous function $\delta > 0$ on X such that $\{|\xi| < \delta(\pi(\xi))\} \subset \Omega$, and put $\bar{s}(\xi) = s(t(\xi)\xi)$, where $t(\xi) = \min\{1, \delta(\pi(\xi))/(2|\xi|)\}$ for $\xi \neq 0$ and $t = 1$ on A . This is a continuous map $E \rightarrow X$ that equals s on $U_0 = \{|\xi| < \delta(\pi(\xi))/2\}$. The derivative of s along E_y at 0_y is $(d^\vee s)_y$, which maps onto $T_{X,y}$. Hence Lemma 3.9, applied with $Y = X$ and $g = s|_{U_0}$, gives a neighbourhood $U' \subset U_0$ of A and a product spray with core $\bar{s}|_{U'}$ that is dominating over $U' \setminus A$.

Conclusion. Since X is Oka, Theorem 3.1, applied with $Y = X$, $f = \bar{s}$, $U = U'$ and $r = 2$, gives a holomorphic map $\tilde{s}: E \rightarrow X$ that agrees with $\bar{s}$ to order 2 along A . Hence

$\tilde{s}(0_y) = s(0_y) = y$, and the differential of $\tilde{s}$ at 0_y coincides with that of s . In particular, $d^\vee \tilde{s} = d^\vee s$ maps onto T_X , so $\tilde{s}$ is a dominating spray on E . $\square$

Proof of Theorem 1.1. Let $H, B = K_X \otimes H^{\otimes(2n+1)}$ and $E = (B^\vee)^{\oplus m}$ be as in the proof of Theorem 4.1. By Claim 4.2, the line bundle B is positive, and by the proof of Theorem 4.1, E admits a local dominating spray.

Since $K_X^{-1} \otimes B \cong H^{\otimes(2n+1)}$ admits a metric with semipositive curvature, induced by the Fubini–Study metric, Theorem 1.4, applied with $k = 1$ and $N = m$, shows that $E = (B^\vee)^{\oplus m}$ admits a dominating spray. $\square$

Corollary 4.3. *Stein Oka manifolds and projective Oka manifolds are elliptic.*

Proof. A Stein manifold is weakly pseudoconvex, and the trivial line bundle with the metric $e^{-\psi}$, for a smooth strictly psh exhaustion function ψ , is positive. A projective manifold is compact, hence weakly pseudoconvex, and it has a positive line bundle. A point is elliptic. Hence Theorem 1.1 applies. $\square$

Both statements are known: the Stein case is classical [8, Chap. 5], and the projective case is due to Forstnerič and Lárusson [10].

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FACULTY OF MATHEMATICS, KYUSHU UNIVERSITY, FUKUOKA 819-0395, JAPAN
 Email address: kusakabe@math.kyushu-u.ac.jp

MATHEMATICAL INSTITUTE & DIVISION FOR THE ESTABLISHMENT OF FRONTIER SCIENCE OF ORGANIZATION FOR ADVANCED STUDIES, TOHOKU UNIVERSITY, 6-3, ARAMAKI AZA-AOBA, AOBA-KU, SENDAI 980-8578, JAPAN
 Email address: mshinichi-math@tohoku.ac.jp
 Email address: mshinichi0@gmail.com